Matter, Then On Down

One coordinate from bulk matter to the nucleus

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Abstract

Matter has no absolute size. The one place scale returns is a logarithmic boundary count whose coefficient is fixed by the central charge, $A = 2c/\pi^2$ for an interval. This paper follows that boundary downward. At every scale the same function, $\Gamma = (z-1)/(z+1) = \tanh(\frac{1}{2}\ln z)$, reads a ratio: occupation odds at a cut of bulk matter, successive removal energies of an atom, separation energies of a nucleus, and the complex odds of opened, non-Hermitian matter. The boundary census is the gauge without its sign; the signed offset $\tau = \log\det(I-C) - \log\det C$ is its clock face, and it moves only as fast as the coupling energy across the cut allows. The thermal entanglement ladder of the Ising chain is a peel ladder with one constant step. Helium's two thresholds give $\Gamma_0 = 0.377575$. The unbound nuclei ^{5}He, ^{5}Li and ^{8}Be sit outside the disk at $|\Gamma| = 1.074$, 1.220, 1.025, recomputed here from NUBASE2020, and their rapidity has turned half a revolution. Opened matter leaves the disk before its critical point; on the PT-symmetric chain the exit line is $E(0)E(\pi) = u^2$, equivalently a band-averaged Petermann factor of exactly 2, and the conditions under which that holds are stated with their counterexamples. Every claim carries a grade and every number a script. Appendix A proves every result the argument uses, including those imported from the author's earlier papers; Appendix B prints the Lean file that machine-checks the algebraic core; Appendix C prints every script, figure scripts included, with its output.

Grades. [T] proved here or in the cited paper of the author; [V] verified numerically with stated controls; [R] a reproduction of a known result in this coordinate; [C] conjecture; [H] a reading. [lit] marks comparison with other people's published data.

1 Matter has no absolute size

Every instrument needs a zero and a span. In the operator framework of *Matter at a Scale* (v5) and *Foundational Operator Field Theory*, the zero is free and so is the span: difference observables are unchanged by $K \mapsto K + cI$, and spectral projectors, principal angles and the infrared boundary census are unchanged by $K \mapsto cK$ with $c > 0$ [T]. Only ratios are observable.

The single failure is ultraviolet. For a region A with $|\partial A|$ boundary points in a substrate of correlation length ξ and central charge c, the number N of eigenvalues of the restricted correlation matrix in $[\varepsilon, 1-\varepsilon]$ grows as

$$N = |\partial A|\,\frac{c}{\pi^2}\,\log\frac{\xi}{a}\,\log\frac{1}{\varepsilon} + \dots \tag{1}$$

so a two-endpoint interval carries $A = 2c/\pi^2$ [T]. The coefficient was measured at $c = 1$ as 0.202692 against $2/\pi^2 = 0.202642$, and at $c = \frac{1}{2}$ (the transverse-field Ising chain) as 0.103689, bracketing $1/\pi^2$ from both sides and 41.8 times closer to it than to the $c = 1$ value [V]. *The Angle Between the Rails* derives the same coefficient from the spacing $\Delta = 4\pi K(\tilde{u})/K'(\tilde{u})$ of the entanglement ladder, $N = 4\Lambda/\Delta$ for two cuts [T]. A rescaling of the substrate is registered only as a running, never as a power.

The physical reading of *Matter at a Scale* is the starting point of everything below: matter is a gap wearing a boundary. A gapped substrate keeps a finite, area-law census; gas, the gapless phase, floods it.

2 The boundary keeps a ledger

2.1 Census, offset and one chart

Let C be the one-body correlation matrix of a region, with eigenvalues $\nu \in (0, 1)$. Each level has odds $z = (1-\nu)/\nu$ and a rapidity $\varepsilon = \ln z$. Write $\Gamma = (z-1)/(z+1)$.

Proposition 1 (the bridge)**.** $\Gamma = \tanh(\varepsilon/2) = 1 - 2\nu$*, and the window* $\nu \in [\delta, 1-\delta]$ *is the disk* $|\Gamma| \le 1 - 2\delta$*. The binary entropy integrated over the rapidity is* $\int_{-\infty}^{\infty} H(\nu(\varepsilon))\, d\varepsilon = \pi^2/3$*. [T]*

Proof. With $z = (1-\nu)/\nu$: $z-1 = (1-2\nu)/\nu$ and $z+1 = 1/\nu$, so $\Gamma = 1-2\nu$. Also $(e^{\varepsilon}-1)/(e^{\varepsilon}+1) = \tanh(\varepsilon/2)$. The window statement is $|1-2\nu| \le 1-2\delta$. For the integral, $\nu = 1/(1+e^{\varepsilon})$ and $d\nu = -\nu(1-\nu)\, d\varepsilon$, so $\int H\, d\varepsilon = \int_0^1 H(\nu)/(\nu(1-\nu))\, d\nu = 2\int_0^1 (-\ln \nu)/(1-\nu)\, d\nu = 2\zeta(2) = \pi^2/3$. □

Checked to 1.1×10^{-16} and 6.7×10^{-15} (script `s01`). The identity $1-2\nu = \tanh(\varepsilon/2)$ is standard in entanglement-spectrum work; the reading is the author's. *The Offset Speed Limit* (Appendix B) names the same function four ways at once: the Smith-chart reflection $(z-1)/(z+1)$, the two-level Gibbs population gap $\tanh(\beta\Delta/2)$, the asymmetry $1-2\nu$ of an occupation against its log-odds, and the peel coordinate of VOTKP.

Summed over the levels of the region, the rapidity is the author's signed offset,

$$\tau = \sum_{\nu} \varepsilon_\nu = \mathrm{Tr}[\log(I-C) - \log C] = \log \frac{\det(I-C)}{\det C} = \log \frac{P(N_A = 0)}{P(N_A = |A|)}, \tag{2}$$

the log-odds of the region being empty against full (*The Offset Belongs to the Boundary*, v31, Theorem 2) [T]. The census counts levels inside a window of $\nu(1-\nu)$, and $\nu(1-\nu) = (1-\Gamma^2)/4$ is blind to the sign of Γ. In the author's words: "The offset is the clock face; the census is the gauge without its sign." The census of $\mathrm{diag}(0.2, 0.8)$ and of $\mathrm{diag}(0.2, 0.2)$ is the same $0.16I$; their offsets are 0 and $\log 16$.

2.2 The weight between and the speed of the record

Split the space into the region A and the rest B. Block multiplication gives the exact defect split

$$C_A - C_A^2 = (C - C^2)_{AA} + X_A, \qquad X_A = C_{AB} C_{BA}. \tag{3}$$

For a pure state $C^2 = C$, so $X_A = C_A(1 - C_A)$ and the eigenvalues of X_A are $\nu(1-\nu)$. Then $\Gamma_b = \sqrt{1 - 4\lambda(X_A)} = |1-2\nu|$: the boundary coordinate is the modulus of the census point, not the signed point [T]. And $\mathrm{Tr}\, X_A = \|P_A C (1-P_A)\|_F^2$ is the coupling energy $E_B = \|PH(1-P)\|^2$ of *The Weight Between*, with the correlation matrix in place of the generator [T].

The offset moves only through that coupling (*The Offset Speed Limit*, Theorem 1 and Corollary 3): for a family of projectors with $C = E^\dagger P E$, $0 < C < I$, $g = \frac{1}{2}\|\dot{P}\|_F^2$ and $S = \mathrm{Tr}[(C(I-C))^{-1}]$,

$$\dot{\tau}^2 \le 4gS, \qquad |\dot{\tau}| \le 2\sqrt{E_B S} \ \text{ when } \dot{P} = -i[H,P], \tag{4}$$

with equality exactly along the trajectories the theorem names [T]. If $PH(1-P) = 0$ the offset is frozen: no weight between, no motion of the record, no memory.

2.3 Warm matter: heat moves the gauge, not the cut

At finite temperature $C^2 \neq C$. The first term of (3) is mixing inside the region; only X_A crosses the cut. Counting $\lambda(X_A) \ge \varepsilon(1-\varepsilon)$ instead of $\nu \in [\varepsilon, 1-\varepsilon]$ separates the two. On a ring of 128 sites (antiperiodic, $\mu = 0$, half-ring region, $\varepsilon = 0.1$) the raw count rises from 2 to 24 to 64 as T goes $0 \to 0.5 \to 2$, while the boundary count goes $2 \to 0 \to 0$; disconnected sites $C = 0.3I$ give raw 32, boundary 0 [V] (script `s07`, Figure 1, left). A boundary count of zero means every cross-cut strength is below the threshold, not that the cut is uncorrelated: $\mathrm{Tr}\, X_A = 0.170835$ in the warm uniform ring. The block identity and these numbers are standard finite-dimensional facts and were also recorded in an independent review of 3 November 2026; the identification of X_A with the weight between is the author's.

2.4 Gas

Matter at a Scale §4 makes gas the gapless phase: the census floods as $L \log L$, directions are free and failures are non-modal. On the disk the gas state is a layer that can be varied freely while an atomic record survives it. In an ideal Saha helium gas built on VOTKP's two thresholds, each pair of neighbouring charge states is a disk point that sweeps the whole diameter with temperature and density, while the density-free combinations $\ln(n_0 n_2/n_1^2)$ and $\ln(n_0/(n_2 n_e^2 \lambda_e^6))$ have temperature slopes $-(I_1 - I_0)$ and $I_0 + I_1$, returning $\Gamma_0 = 0.377575066984$ at every density from 10^{18} to $10^{26}\,\mathrm{m}^{-3}$ (Figure 1, right). This is a reproduction in the declared ideal model [R]; with temperature-dependent partition functions the naive read-back is biased (to 0.366149 for a synthetic excited level) and the known correction restores it.

3 One chart for all of it

Theorem 2 (peel steps add like velocities; VOTKP Theorem 15.1)**.** *For a positive ladder $I_0, \dots, I_n$ with steps $\Gamma_k = \tanh(\frac{1}{2}\ln(I_{k+1}/I_k))$, the composite of a block of steps is $\Gamma_{j\to l} = \Gamma_j \oplus \Gamma_{j+1} \oplus \cdots \oplus \Gamma_{l-1}$, with $u \oplus v = (u+v)/(1+uv)$. [T]*

Proof. $\ln(I_l/I_j) = \sum_k \ln(I_{k+1}/I_k)$, and $\tanh(a+b) = \tanh a \oplus \tanh b$. □

This is Einstein's addition of collinear velocities; the half-rapidities add. On the Smith disk read as the Poincaré disk the peel is a walk along a geodesic diameter whose hyperbolic lengths add (VOTKP Corollary 15.2) [T]. The composite forgets the order of its steps; the resolved ladder keeps it.

Matter's census ladder is a peel ladder. On the transverse-field Ising chain ($c = \frac{1}{2}$) the single-particle entanglement energies are odd multiples $\varepsilon_0, 3\varepsilon_0, 5\varepsilon_0, \dots$ of one spacing (*Matter at a Scale*, LADDER-C12-01). By Theorem 2 the levels then satisfy $\Gamma_{j+1} = \Gamma_j \oplus \tanh\varepsilon_0$: one constant peel step. Measured residuals are 8.0×10^{-6}, 5.5×10^{-6}, 3.2×10^{-6} at $\xi = 20.5$, 50.5, 125.5, and

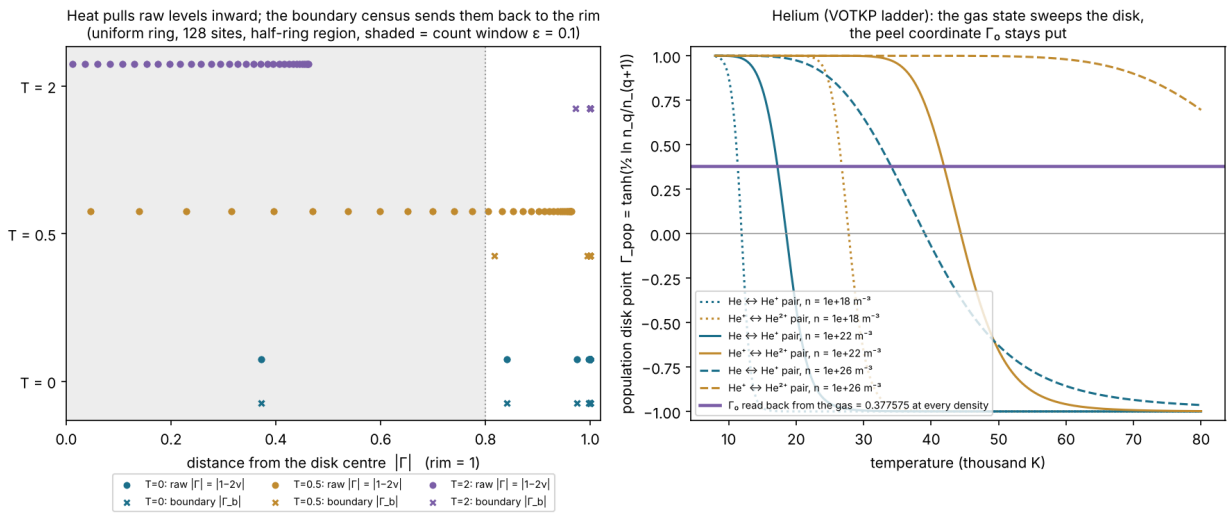


Figure 1: Left: heat pulls raw census levels toward the centre of the disk; the boundary coordinate $|\Gamma_b|$ keeps only what crosses the cut. Right: helium's charge-state pairs sweep the disk as the gas changes, while Γ_0 read back from the gas stays at 0.377575.

the ladder's own uniformity error falls 0.022, 0.011, 0.0062 [V] (script `s02`). Near the rim the disk compresses differences, so the informative number is the uniformity error in rapidity.

4 The atom

For an atom the ratio is between successive relaxed removal thresholds, $z_q = I_{q+1}/I_q$, and every step lies on the positive real radius (VOTKP §8, Theorem 8.1 chart). Helium's two thresholds, 24.587389011 and 54.4177655282 eV, give $\Gamma_0 = 0.377575066984$ and a binding sum of 79.0051545392 eV [T]. An anchor and the full coordinate vector reconstruct the ladder exactly, $I_q = I_0 \prod_{j<q}(1+\Gamma_j)/(1-\Gamma_j)$, and a common scale leaves every Γ unchanged [T]. VOTKP Theorem 15.4 (scale blindness) is the atomic form of §1: multiplying every mass by λ scales every energy by λ and leaves every ratio fixed, so the peel coordinate is blind to which particle carries the charge except through charges, mass ratios and statistics [T]. The coordinate needs a physical zero: removal energies are measured from the continuum, which is why the ratio means something, and why quantum-dot addition spectra, which can be moved up or down, have no peel (VOTKP §15.3).

Sodium's eleven thresholds walk from the centre to $\Gamma = 0.9938$ along one diameter (Figure 2); shell jumps are the long strides. The last peel step is derived in closed form (VOTKP §15.10), and the middle of the peel follows the Thomas–Fermi law with $\Gamma_k \to \tanh(\frac{2}{3}\ln\frac{k+1}{k}) \approx 2/(3k)$ (§15.11, cited there) [T]/[lit].

5 The nucleus

A nucleus peels by separation energies. VOTKP §15.4 uses $z = S_{\text{next}}/S_{\text{first}}$ and the same Γ, "the form that extends to $z < 0$". A bound peel has $z > 0$ and sits inside the disk. A negative separation energy gives $z < 0$ and $|\Gamma| > 1$; $z = -1$ is a Cayley pole, and the real logarithmic rapidity is not extended across it. Recomputed here from NUBASE2020 with the author's own script:

case	S_{first} (keV)	S_{next} (keV)	$\lvert\Gamma\rvert$	half-life
^{5}He: n, then n from ^{4}He	-734.8	20577.6	1.074	602 ys
^{5}Li: p, then p from ^{4}He	-1966.1	19813.9	1.220	370 ys
^{8}Be: α (^{12}C ladder)	-91.8	7366.6	1.025	81.9 as
^{7}Li (bound)	7251.1	5664.4	0.123	stable
^{4}He (bound)	20577.6	5493.5	0.579	stable

These agree with VOTKP §15.4 to every printed digit [V][lit].

How far outside, exactly. VOTKP §15.4 notes that the distance outside the rim is close to twice the decay energy over the neighbouring step. It is exact in this form. Write $S_{\text{first}} = -s < 0$ and $a = S_{\text{next}}/s > 1$, so $z = -a$ and

$$|\Gamma| = \frac{a+1}{a-1}, \qquad |\Gamma| - 1 = \frac{2}{a-1} = \frac{2s}{S_{\text{next}} - s},$$

which is $2s/S_{\text{next}}$ when $s \ll S_{\text{next}}$ [T]. On the three gap nuclei the identity holds exactly (0.074062, 0.220318, 0.025238), against the small-s values 0.071417, 0.198457, 0.024923 (script `s12`). The Cayley pole $z = -1$ is $S_{\text{first}} + S_{\text{next}} = 0$. The placement says no more than the sign of a separation energy, and the author's own caution stands: a resonance's leak through its barrier "is not a literal atomic or nuclear Smith reflection derived from the threshold coordinate".

The half-turn. Write $\Gamma = \tanh(W/2)$ with $W = \operatorname{Log} z$ (principal branch). For $z \neq -1$, $|z+1|^2 - |z-1|^2 = 4\operatorname{Re} z$, so $|\Gamma| < 1 \iff \operatorname{Re} z > 0 \iff |\operatorname{Im} W| < \pi/2$ [T]. A negative ratio has $\operatorname{Im} W = \pi$ exactly: the three unbound nuclei are peels whose rapidity has turned half a revolution, and the rim of the disk is the quarter-turn. Random complex z (4000 draws) give zero mismatches, and $z = -28$ gives $\Gamma = 1.0741$, the ^{5}He value [V] (script `s04a`). A complex ratio by itself does not leave the disk: $z = 1 - i\gamma$ stays inside for every γ, since $\operatorname{Re} z > 0$. Leaving needs a ratio whose real part changes sign.

6 Opened matter

6.1 The opened chain

Opening matter means a non-Hermitian generator. The cone paper (v8) proves that at a Hermitian point the round-trip form $Q(\theta) = M_{+-}M_{-+}/(E_+ - E_-)^2$ has $\operatorname{Re} Q \geq 0$, so no cone exists there, and measures a Lorentzian cone when the system is opened [T]/[V]. Take the PT-symmetric SSH chain

$$h(k) = \begin{pmatrix} iu & v + we^{-ik} \\ v + we^{ik} & -iu \end{pmatrix}, \qquad E(k)^2 = |v + we^{ik}|^2 - u^2,$$

fill the lower band with the biorthogonal projector, and restrict to a block. The occupations are complex, and so is $\Gamma = 1 - 2\nu$; the biorthogonal C is still idempotent ($\|C^2 - C\| \leq 3 \times 10^{-14}$), so $Y_A = C_{AB}C_{BA} = C_A - C_A^2$ and the radial census $\sqrt{|1 - 4\lambda|} = |1 - 2\nu|$ survive opening [T][V]. On the opened plane at $u = 0.5$ the band metric is Lorentzian over 93% of the cells, its determinant changes sign exactly at the critical points $|v - w| = u$, and the closed plane is degenerate [V]. At the critical point the Rényi-2 central charge reads $-1.79 \to -1.96$ on the $v < w$ side [V]; the $v > w$ side has not converged. A negative fitted coefficient is a property of the biorthogonal entropy convention, not a negative number of channels.

6.2 Where opened matter leaves the disk

Approaching the critical point at fixed u, the census levels leave the disk *before* it, inside the gapped phase, and the exit does not move under a fourfold larger block or a doubled ring ($v_{\text{exit}} = 0.468$ at $u = 0.5$) [V]. At every critical point tested ($u = 0.2, \ldots, 0.7$) a level sits at $\operatorname{Im} W = \pi$ exactly: the half-turn of the unbound nuclei.

Proposition 3 (the exit line of the PT-symmetric chain)**.** *The Petermann factor of the lower band is $K(k) = 1 + u^2/E(k)^2$, and for $A = v^2 + w^2 - u^2 > |2vw|$ its band average is*

$$\langle K \rangle = 1 + \frac{u^2}{\sqrt{A^2 - B^2}} = 1 + \frac{u^2}{E(0)E(\pi)}, \qquad B = 2vw.$$

Hence $\langle K \rangle = 2 \iff E(0)E(\pi) = u^2$. [T]

Proof. For $h = d \cdot \sigma$ with $E^2 = d \cdot d$ the band projector is $P = \frac{1}{2}(I - h/E)$ and $K = \operatorname{Tr} P^\dagger P = \frac{1}{2}(1 + \sum |d_j|^2/|E|^2)$. Here $d = (\operatorname{Re} q, \operatorname{Im} q, iu)$, so $\sum |d_j|^2 = |q|^2 + u^2$ and $|E|^2 = |q|^2 - u^2$, giving $K = 1 + u^2/E^2$. With $E^2 = A + B\cos k$ and $\frac{1}{2\pi}\int_0^{2\pi} \frac{dk}{A + B\cos k} = \frac{1}{\sqrt{A^2 - B^2}}$, and $\sqrt{A^2 - B^2} = \sqrt{A + B}\sqrt{A - B} = E(0)E(\pi)$. □

Checked to 3.6×10^{-14}. Predicted from $\langle K \rangle = 2$ at three values of u not used before, the exits are 0.7474014, 0.5291503, 0.2357262; measured, 0.7474015, 0.5291503, 0.2357262 (differences 4×10^{-8} and below) [V].

The limits of the exit law. Three findings scope it, and they belong in the statement.

1. On the cone paper's band made one-dimensional, $d = (t_1 + \cos k, \sin k, m + i\gamma)$, a mass term moves the exit far earlier ($\langle K \rangle \approx 1.015$–$1.022$); the law holds only at $m = 0$, which is the SSH class [V].

2. The massive exit depends on the endpoint filter that discards levels with ν at 0 or 1: $\gamma_{\text{exit}} = 0.27164$ at filters 10^{-6} and 10^{-8}, 0.16917 at 10^{-9} and 10^{-10}, 0.12272 at 10^{-12}, while the massless exit stays at 0.47434 through 10^{-8} [V]. This was found in an independent formal review (3 November 2026) and reproduced here (script `s09`).

3. With $w = 0$ and $v^2 = 2u^2$ the band average is exactly 2, yet no block level ever leaves $\{0, 1\}$ [T]. So the average alone does not force an exit; the coupling and the partition are part of the law.

What survives: on the coupled PT-symmetric chain with whole-cell blocks and the stated filter, the census leaves the disk where $E(0)E(\pi) = u^2$, to 4×10^{-8} [V]; why it does is open [C].

6.3 The offset as the exit's clock

For opened matter each level's offset $\tau_\nu = \operatorname{Log}((1 - \nu)/\nu)$ is complex, and the rotation used above is $|\operatorname{Im} \tau_\nu|/\pi$. The disk exit is the place where some level's signed offset has turned a quarter, and the critical point the place where one has turned a half.

The author's closed form for the endpoint offset of a gapped chain, $2\operatorname{artanh}(v/\sqrt{E_{\min}E_{\max}})$ (*The Offset Belongs to the Boundary*), continued to the opened chain, has imaginary part $\operatorname{Im} \tau = 2\arctan\left(u/\sqrt{E(0)E(\pi)}\right)$. Then exactly

$$\langle K \rangle = 1 + \frac{u^2}{E(0)E(\pi)} = \sec^2\Big(\tfrac{1}{2}\operatorname{Im} \tau\Big),$$

so $\langle K\rangle = 2$ is the same statement as $\operatorname{Im}\tau = \pi/2$: the exit law of §6.2 is a quarter-turn of this offset [T]. It is not, however, a description of the census rotation away from the exit. Tested before the run with a kill condition, at $u = 0.3, 0.5, 0.7$ and three points on the approach, the measured rotation sits 4×10^{-6} to 3.4×10^{-2} above the formula, independent of block size; the two meet only at the quarter-turn. The level that carries the largest rotation is often one with ν within 10^{-7} of an endpoint, the same ill-conditioned levels as the filter result above [V]. Whether a biorthogonal form of the speed limit $\dot\tau^2 \le 4gS$ governs the rotation is open [C].

7 What is proved, what is measured, what is open

Statement	grade
Zero and span free; census logarithm with $A = 2c/\pi^2$	[T], [V] at $c = 1, \frac{1}{2}$
$\Gamma = \tanh(\varepsilon/2) = 1 - 2\nu$; $\int H\, d\varepsilon = \pi^2/3$	[T]
Offset = summed level rapidity = log-odds empty/full	[T]
$\Gamma_b = \lvert 1 - 2\nu\rvert$; $\operatorname{Tr} X_A$ = the weight between	[T]
Offset speed limit $\dot\tau^2 \le 4gS$	[T]
Warm census: heat moves raw levels, not boundary levels (ring, 128 sites)	[V]
Ising census ladder = constant peel step	[V]
Peel steps add like velocities	[T]
Ideal helium gas returns Γ_0 at every density	[R]
Unbound nuclei outside the disk; half-turn of the rapidity	[T], [V][lit]
Rim distance $\lvert\Gamma\rvert - 1 = 2s/(S_{\text{next}} - s)$	[T]
$\langle K\rangle = \sec^2(\frac{1}{2}\operatorname{Im}\tau)$: exit = quarter-turn of the offset	[T]
Biorthogonal idempotence; radial census survives opening	[T], [V]
Opened matter exits before critical; exit line $E(0)E(\pi) = u^2$ (PT class, stated filter and partition)	[V]
$\langle K\rangle = 1 + u^2/(E(0)E(\pi))$	[T]

Open: why the census rotation meets the offset quarter-turn only at the exit, which needs conditioned eigenvalues for the near-endpoint levels; a proof that the exit happens exactly at $\langle K\rangle = 2$ on the coupled chain; a filter-independent definition of the massive exit (filter, ring and block sizes, order of limits); a biorthogonal speed limit for the complex offset; convergence on the $v > w$ side; placing nuclear resonances with their widths on the chart; a second non-unitary model of a different kind.

Sources

J. Beasley, *Matter at a Scale: Zero, Span, and the Logarithm Between Them*, v5 (2026), doi:10.5281/zenodo.22019893. J. Beasley, *VOTKP: The Vector of Traits Kept under Peeling*, v2.9 (2026), doi:10.5281/zenodo.23340757. J. Beasley, *The Offset Belongs to the Boundary*, v31 (2026), doi:10.5281/zenodo.22555919. J. Beasley, *The Offset Speed Limit*, v1.1 (2026). J. Beasley, *The Weight Between*, v2.3 (2026). J. Beasley, *The Angle Between the Rails* (2026). J. Beasley, *A Measured Lorentzian Cone in Non-Hermitian Quantum Geometry*, v8 (2026). J. Beasley, *Beyond the Event Horizon*, edition 10 (2026). F. G. Kondev et al., The NUBASE2020 evaluation of nuclear physics properties, Chin. Phys. C 45 (2021) 030001. A. Kramida et al., NIST Atomic Spectra Database (ionization energies), accessed through the VOTKP v2.9 package. P.-Y. Chang, J.-S. You, X. Wen, S. Ryu, Phys. Rev. Research 2 (2020) 033069. Independent formal review, *Matter, Gases and VOTKP*, 3 November 2026 (prepared with OpenAI tools), kept in the repository.

A Proofs of every result used

This appendix makes the paper standalone. Each result used in the text is restated and proved here, including those taken from the author's earlier papers. Where a proof rests on an outside input, the input is named and is the only thing assumed. Items marked [Lean] are also checked by machine (Appendix B).

A.1 Zero and span are free (§1). *Statement.* For a Hermitian generator K, a shift $K \mapsto K + cI$ leaves every eigenvector and every eigenvalue difference unchanged, and a positive rescaling $K \mapsto cK$, $c > 0$, leaves every eigenvector, the order of the eigenvalues, every spectral projector $\mathbf{1}[K < 0]$, and so the ground-state correlation matrix and its census, unchanged. *Proof.* $K + cI$ commutes with K and has eigenvalues $\lambda_i + c$, so $\lambda_i - \lambda_j$ is unchanged. cK has the same eigenvectors with eigenvalues $c\lambda_i$; since $c > 0$, $c\lambda_i < 0 \iff \lambda_i < 0$, so the filled projector $P = \mathbf{1}[K < 0]$ is the same operator. The regional correlation matrix $C_A = E^\dagger P E$ and every function of its spectrum, including the census, are therefore unchanged; so are principal angles between spectral subspaces, which depend only on the subspaces. □

A.2 The census coefficient $A = 2c/\pi^2$ (§1, eq. 1). *Inputs.* (i) The single-particle entanglement spectrum at one cut is equally spaced, $\varepsilon_j = (j + \frac{1}{2})\Delta$ on both signs, as measured on the Ising chain (§3, script `s02`) and derived in *The Angle Between the Rails*. (ii) The entanglement entropy of one cut in a gapped chain near criticality is $S_{\text{cut}} = \frac{c}{6}\ln(\xi/a) + O(1)$ (Calabrese–Cardy). *Derivation.* A level is in the window $\nu \in [\epsilon, 1-\epsilon]$ iff $|\varepsilon| \le \Lambda = \ln((1-\epsilon)/\epsilon) = \ln(1/\epsilon) + O(\epsilon)$ (Proposition 1). With spacing Δ on both signs, one cut has $N_{\text{cut}} = 2\Lambda/\Delta + O(1)$ levels in the window. Its entropy is $S_{\text{cut}} = \sum_j H(\nu(\varepsilon_j)) = \Delta^{-1}\int H\, d\varepsilon + O(\Delta) = \pi^2/(3\Delta) + O(\Delta)$ by Proposition 1 (midpoint rule). Equating with (ii), $1/\Delta = (c/2\pi^2)\ln(\xi/a)$, so $N_{\text{cut}} = (c/\pi^2)\ln(\xi/a)\ln(1/\epsilon)$, and an interval with two cuts has $A = 2c/\pi^2$. □ The derivation is exact given (i) and (ii); its corrections are those of the two inputs.

A.3 The bridge, and $\int H\, d\varepsilon = \pi^2/3$ (Proposition 1). Proved in the text. The last step uses $\int_0^1 (-\ln\nu)/(1-\nu)\, d\nu = \sum_{n\ge 0}\int_0^1 \nu^n(-\ln\nu)\, d\nu = \sum_{n\ge 1} n^{-2} = \pi^2/6$, and $\int_0^1 H/(\nu(1-\nu))\, d\nu = 2\int_0^1 (-\ln\nu)/(1-\nu)\, d\nu$ by the symmetry $\nu \leftrightarrow 1-\nu$ and $H/(\nu(1-\nu)) = -\ln\nu/(1-\nu) - \ln(1-\nu)/\nu$. The identities $\Gamma = 1 - 2\nu$ and $\nu(1-\nu) = (1-\Gamma^2)/4$ [Lean: `bridge`, `census_square`].

A.4 The offset is the log-odds of empty against full (§2.1; *The Offset Belongs to the Boundary*, Thm 2). *Statement.* For a number-conserving Gaussian fermion state with $0 < C_A < I$, $\tau = \mathrm{Tr}[\log(I - C_A) - \log C_A] = \log\det(I - C_A) - \log\det C_A = \log P(N_A = 0)/P(N_A = |A|)$. *Proof.* The reduced state of a Gaussian state is Gaussian, $\rho_A \propto \exp(-\sum_k \varepsilon_k b_k^\dagger b_k)$ in the eigenmodes b_k of C_A, so it is a product over modes with occupation $\nu_k = 1/(1 + e^{\varepsilon_k})$. The region is empty iff every mode is empty and full iff every mode is filled, so $P(0) = \prod_k (1 - \nu_k) = \det(I - C_A)$ and $P(|A|) = \prod_k \nu_k = \det C_A$. Taking logs, $\tau = \sum_k \ln((1-\nu_k)/\nu_k) = \sum_k \varepsilon_k$. The example in the text: $\mathrm{diag}(0.2, 0.8)$ gives $\ln 4 + \ln\frac{1}{4} = 0$, $\mathrm{diag}(0.2, 0.2)$ gives $2\ln 4 = \ln 16$, while both have $C - C^2 = 0.16I$. □

A.5 Defect split and the weight between (§2.2). *Statement.* Write $C = \begin{pmatrix} C_A & C_{AB} \\ C_{BA} & C_B \end{pmatrix}$. Then $(C^2)_{AA} = C_A^2 + C_{AB}C_{BA}$, so $C_A - C_A^2 = (C - C^2)_{AA} + X_A$ with $X_A = C_{AB}C_{BA}$. If $C^2 = C$ then $X_A = C_A - C_A^2$, with no Hermiticity used. If C is Hermitian, $\mathrm{Tr}\, X_A = \|C_{AB}\|_F^2 = \|P_A C(1 - P_A)\|_F^2$.

Proof. Block multiplication gives the first identity [Lean: `topleft_sq`]; subtracting it from C_A gives the split. If $C^2 = C$ the first term vanishes [Lean: `idempotent_cross`]. For Hermitian C, $C_{BA} = C_{AB}^\dagger$, so $\operatorname{Tr} C_{AB}C_{AB}^\dagger = \|C_{AB}\|_F^2$, and $P_AC(1-P_A)$ is C_{AB} padded with zeros. With H in place of C this is the coupling energy $E_B = \|PH(1-P)\|_F^2$ of *The Weight Between.* □

A.6 The boundary coordinate is the modulus (§2.2), and it survives opening (§6.1). *Statement.* If $C^2 = C$, each eigenvalue λ of X_A is $\nu(1-\nu)$ for an eigenvalue ν of C_A, and $\sqrt{|1-4\lambda|} = |1-2\nu|$. This holds for the biorthogonal projector $C = \sum_{k\,\text{filled}} |R_k\rangle\langle L_k|$ with $\langle L_j|R_k\rangle = \delta_{jk}$. *Proof.* By A.5, $X_A = C_A - C_A^2 = f(C_A)$ with $f(x) = x - x^2$, so by the spectral mapping theorem (valid for any square matrix, via the Jordan form) its eigenvalues are $f(\nu)$. Then $1 - 4f(\nu) = (1-2\nu)^2$ [Lean: `census_square`], whose modulus has square root $|1-2\nu|$ for complex ν too. For the biorthogonal projector, $C^2 = \sum_{j,k} |R_j\rangle\langle L_j|R_k\rangle\langle L_k| = C$. □

A.7 The Offset Speed Limit (§2.2; *The Offset Speed Limit*, Thm 1 and Cor. 3). *Statement.* Let $P(t)$ be a smooth family of orthogonal projectors on $\mathbb{C}^n$, E an $n \times m$ isometry onto the region, $C = E^\dagger PE$ with $0 < C < I$, $\tau = \operatorname{Tr}[\log(I-C) - \log C]$, $g = \frac{1}{2}\|\dot P\|_F^2$ and $S = \operatorname{Tr}[(C(I-C))^{-1}]$. Then $\dot\tau^2 \le 4gS$, with equality iff $\dot P$ is a real multiple of $T = [P,[P,F]]$, $F = EWE^\dagger$, $W = -(C(I-C))^{-1}$. If $\dot P = -i[H,P]$ then $g = E_B$ and $|\dot\tau| \le 2\sqrt{E_B S}$. *Proof. Step 1 (trace derivative).* $\frac{d}{dt}\operatorname{Tr}\log(I-C) = -\operatorname{Tr}[(I-C)^{-1}\dot C]$ and $\frac{d}{dt}\operatorname{Tr}\log C = \operatorname{Tr}[C^{-1}\dot C]$; since $(I-C)^{-1} + C^{-1} = (C(I-C))^{-1}$, $\dot\tau = \operatorname{Tr}[W\dot C]$. *Step 2 (lift).* $\dot C = E^\dagger \dot P E$, so $\dot\tau = \operatorname{Tr}[EWE^\dagger\dot P] = \operatorname{Tr}[F\dot P]$, with F Hermitian. *Step 3 (only the off-diagonal part counts).* Differentiating $P^2 = P$ gives $\dot P P + P\dot P = \dot P$, hence $P\dot P P = 0$ and $(1-P)\dot P(1-P) = 0$: $\dot P$ is block off-diagonal. So $\operatorname{Tr}[F\dot P] = \operatorname{Tr}[F_{\text{off}}\dot P]$ with $F_{\text{off}} = PF(1-P) + (1-P)FP$, and expanding, $[P,[P,F]] = PF + FP - 2PFP = F_{\text{off}} = T$. *Step 4 (Cauchy–Schwarz).* In the Hilbert–Schmidt inner product, $\dot\tau^2 = \langle T, \dot P\rangle^2 \le \|T\|_F^2\|\dot P\|_F^2 = 2g\|T\|_F^2$, with equality iff $\dot P = \lambda T$, $\lambda \in \mathbb{R}$. *Step 5 ($\|T\|_F^2 = 2S$).* $\|T\|_F^2 = 2\operatorname{Tr}[PF(1-P)FP] = 2\operatorname{Tr}[W\,E^\dagger(1-P)E\,W\,E^\dagger PE] = 2\operatorname{Tr}[W(I-C)WC] = 2\operatorname{Tr}[W^2C(I-C)] = 2S$, because W, C and $I-C$ commute. Hence $\dot\tau^2 \le 4gS$. For the corollary, $[H,P] = (1-P)HP - PH(1-P)$, whose two blocks are adjoint up to sign, so $\|\dot P\|_F^2 = 2\|PH(1-P)\|_F^2$ and $g = E_B$. □

A.8 Peel steps add like velocities (Theorem 2; VOTKP Thm 15.1). Proved in the text. In the division-free form used by the Lean file, $g(z_1) + g(z_2) = g(z_1z_2)\,(1 + g(z_1)g(z_2))$ for $z_1, z_2 > 0$, with $g(z) = (z-1)/(z+1)$ [Lean: `rapidities_add`]. Since $1 + g_1g_2 > 0$ for $|g_i| < 1$, this is $g(z_1z_2) = g_1 \oplus g_2$. The ladder reconstruction $I_q = I_0 \prod_{j<q}(1+\Gamma_j)/(1-\Gamma_j)$ is the inverse $z = (1+\Gamma)/(1-\Gamma)$ applied step by step.

A.9 Scale blindness (§4; VOTKP Thm 15.4). *Statement.* For a nonrelativistic Coulomb system $H_\lambda = \sum_i p_i^2/(2\lambda m_i) + \sum_{i<j} q_iq_j/|r_i - r_j|$, multiplying every mass by $\lambda > 0$ multiplies every energy by λ, so every ratio of removal energies, and every Γ, is unchanged. *Proof.* Let U be the unitary dilation $(U\psi)(r) = \lambda^{3N/2}\psi(\lambda r)$. Then $U^\dagger r_i U = r_i/\lambda$ and $U^\dagger p_i U = \lambda p_i$, so $U^\dagger H_\lambda U = \sum_i \lambda^2 p_i^2/(2\lambda m_i) + \lambda\sum_{i<j} q_iq_j/|r_i - r_j| = \lambda H_1$. Unitarily equivalent operators have the same spectrum, so $\operatorname{spec} H_\lambda = \lambda \operatorname{spec} H_1$, and so do the thresholds of every ion, which are bottoms of spectra of subsystems. Ratios cancel λ. □ Mass *ratios* still enter, which is why the ^{1}H/^{2}H isotope shift is visible (VOTKP §15.3).

A.10 The last peel step at first order (§4; VOTKP §15.10). *Statement.* To first order in the electron repulsion, a helium-like ion of charge Z has removal threshold $I_{\text{He}} = Z^2/2 - 5Z/8$

Hartree, the hydrogen-like ion $I_{\mathrm{H}} = Z^2/2$, and the last step has $\Gamma = 5/(8Z-5) \to 5/(8Z)$. *Proof.* The hydrogen-like ground energy is $-Z^2/2$. Two non-interacting $1s$ electrons have $-Z^2$. The first-order repulsion is $J = \iint \rho(r_1)\rho(r_2)/|r_1 - r_2|$ with $\rho = (Z^3/\pi)e^{-2Zr}$. By Gauss's law the potential of ρ is $\phi(r) = 1/r - (Z + 1/r)e^{-2Zr}$, so

$$J = 4Z^3 \int_0^\infty r^2 e^{-2Zr} \phi(r)\, dr = 4Z^3 \Big[\tfrac{1}{4Z^2} - \tfrac{2Z}{(4Z)^3} - \tfrac{1}{(4Z)^2} \Big] = 4Z^3 \cdot \tfrac{10}{64Z^2} = \tfrac{5}{8} Z$$

(Unsöld 1927). So $E_{\mathrm{He}} = -Z^2 + 5Z/8$ and $I_{\mathrm{He}} = -Z^2/2 - E_{\mathrm{He}} = Z^2/2 - 5Z/8$. With $z = I_{\mathrm{H}}/I_{\mathrm{He}}$, $\Gamma = (I_{\mathrm{H}} - I_{\mathrm{He}})/(I_{\mathrm{H}} + I_{\mathrm{He}}) = (5Z/8)/(Z^2 - 5Z/8) = 5/(8Z-5)$ [Lean: `last_step`]. □ The higher terms of the $1/Z$ series (Hylleraas; Scherr–Knight) are cited inputs in VOTKP and are not used here. The Thomas–Fermi middle of the peel (§15.11) is cited there as an external result and is not proved in this paper.

A.11 Ideal Saha gas: the electron density cancels (§2.4). *Input.* The ideal Saha equation $n_{i+1} n_e / n_i = (2g_{i+1}/g_i)\lambda_e^{-3} e^{-\beta I_i}$ with constant statistical weights. *Statement.* $n_0 n_2 / n_1^2 = (g_0 g_2 / g_1^2)\, e^{-\beta(I_1 - I_0)}$ and $n_0/(n_2 n_e^2 \lambda_e^6) = (g_0/4g_2)\, e^{\beta(I_0 + I_1)}$, independent of density; their β-slopes return $I_1 - I_0$ and $I_0 + I_1$, hence $\Gamma_0 = (I_1 - I_0)/(I_1 + I_0)$. *Proof.* Divide the $i = 1$ equation by the $i = 0$ one: n_e and λ_e cancel [Lean: `saha_ratio, the cancellation step`]. Multiply the two equations for the second form. The ratio of the two slopes gives Γ_0. □ With temperature-dependent partition functions the slopes acquire $d \ln g / d\beta$ terms, which is the bias stated in §2.4.

A.12 Outside the rim, exactly, and the Cayley pole (§5). For $S > s > 0$ and $z = S/(-s)$: $\Gamma = (S+s)/(S-s)$ and $\Gamma - 1 = 2s/(S-s)$; $S_2/S_1 = -1 \iff S_1 + S_2 = 0$ [Lean: `outside_rim`, `rim_distance`, `cayley_pole`]. The half-turn statement $|\Gamma| < 1 \iff \mathrm{Re}\, z > 0$ is proved in the text from $|z+1|^2 - |z-1|^2 = 4\,\mathrm{Re}\, z$; and $\mathrm{Re}\, z > 0 \iff |\,\mathrm{Im}\,\mathrm{Log}\, z| < \pi/2$ because $\mathrm{Re}\, z = |z| \cos(\mathrm{Im}\,\mathrm{Log}\, z)$.

A.13 No cone at a Hermitian point (§6.1; cone paper v8, Thm 1). *Statement.* For a Hermitian two-level $h(\theta)$ with $E_+ \neq E_-$, $Q = M_{+-}M_{-+}/(E_+ - E_-)^2 \geq 0$, where $M_{ab} = \langle L_a | \partial_\theta h | R_b \rangle$. *Proof.* At a Hermitian point $L_a = R_a$ and $\partial_\theta h$ is Hermitian, so $M_{-+} = \overline{M_{+-}}$ and $E_\pm$ are real. Hence $Q = |M_{+-}|^2/(E_+ - E_-)^2 \geq 0$. □

A.14 The Petermann factor and the exit line (§6.2). The formula $K = \frac{1}{2}(1 + \sum |d_j|^2/|E|^2)$ follows from $P = \frac{1}{2}(I - h/E)$ and $\mathrm{Tr}\, \sigma_j \sigma_k = 2\delta_{jk}$: $\mathrm{Tr}\, P^\dagger P = \frac{1}{4}\,\mathrm{Tr}[(I - \bar{d}\cdot\sigma/\bar{E})(I - d\cdot\sigma/E)] = \frac{1}{2}(1 + \bar{d}\cdot d/|E|^2)$, the cross terms being traceless. The rest is proved in the text; the final equivalence $1 + u^2/p = 2 \iff p = u^2$ for $p > 0$ [Lean: `exit_line`]. The integral $\frac{1}{2\pi}\int_0^{2\pi} dk/(A + B\cos k) = (A^2 - B^2)^{-1/2}$ for $A > |B|$ is the residue at the root of $Bz^2 + 2Az + B$ inside the unit circle.

A.15 The quarter-turn identity (§6.3). Let $p = E(0)E(\pi) > 0$ and define $\theta = 2\arctan(u/\sqrt{p}) \in (0, \pi)$, the imaginary part of the author's closed form continued to the opened chain (that continuation is a definition, and §6.3 tests it and finds it does not describe the rotation away from the exit). Then $\tan^2(\theta/2) = u^2/p$, so $\sec^2(\theta/2) = 1 + u^2/p = \langle K \rangle$, and $\langle K \rangle = 2 \iff \tan(\theta/2) = 1 \iff \theta = \pi/2$. □

What this appendix does not prove. The numerical statements graded [V] are measurements, and their evidence is the scripts of Appendix C. The two inputs of A.2, the Saha equation of A.11, the $1/Z$ coefficients and the Thomas–Fermi law cited in VOTKP, and the NUBASE2020 and NIST

data are outside inputs. Why the census leaves the disk exactly at $\langle K \rangle = 2$ on the coupled chain is open, as stated in §7.

B The Lean file

The algebraic core is checked in Lean 4 with Mathlib (toolchain and Mathlib v4.34.1), module `VOTKP.MatterDisk` of the repository `dicipler-pixel/votkp-lean`. Its twelve theorems are: `bridge`, `census_square`, `census_complement` (A.3, A.6); `rapidities_add` (A.8); `topleft_sq`, `idempotent_cross` (A.5, A.6); `exit_line` (A.14); `outside_rim`, `rim_distance`, `cayley_pole` (A.12); `saha_ratio` (A.11); `last_step` (A.10).

Status: pending the CI run.

The analytic results (A.2, A.4, A.7, A.9, the integrals of A.3, A.10 and A.14) are proved by hand above and are not machine-checked.

```
/-
Matter, Then On Down (Jeromie Beasley, draft v3, 3 November 2026).
The algebraic core of the paper, machine-checked.

* The bridge: for an occupation `0 < ν< 1` with odds `z = (1 -ν)/ν`, `(z -1)/(z + 1) = 1 -2ν`.
* The census is the gauge without its sign: `ν(1 -ν) = (1 -(1 -2ν)^2)/4`.
* Peel steps add like velocities: with `g z = (z -1)/(z + 1)`,
  `g z₁ + g z₂ = g (z₁ z₂) ·(1 + g z₁ ·g z₂)` for `z₁, z₂ > 0`.
* Defect split: the top-left block of `C^2` is `A^2 + B E`, so `A -A^2 = (A -(C^2)₁₁) + B E`.
* Idempotence survives opening: if `C^2 = C` then `B E = A -A^2`, with no Hermiticity assumed.
* The exit line: for `p > 0`, `1 + u^2/p = 2 ↔p = u^2`.
* The nucleus: for `S > s > 0` and `z = S/(-s)`, `Γ = (S + s)/(S -s)`, so `Γ -1 = 2s/(S -s)`;
  the Cayley pole `z = -1` is `S_first + S_next = 0`.
* The ideal gas: two Saha ratios give `n₀ n₂ / n₁^2 = S₁/S₀` with the electron density cancelled.
* The last peel step at first order: `z = (Z^2/2)/(Z^2/2 -5Z/8)` gives `Γ = 5/(8Z -5)`.
-/
import Mathlib

namespace VOTKP.MatterDisk

/-- The Smith coordinate of a ratio. -/
noncomputable def g (z : ℝ) : ℝ:= (z - 1) / (z + 1)

/-- **The bridge.** `Γ = 1 -2ν` for the odds `z = (1 -ν)/ν`. -/
theorem bridge (ν : ℝ) (h0 : 0 < ν) (h1 : ν< 1) : g ((1 - ν) / ν) = 1 - 2 * ν:= by
  have hν : ν≠0 := h0.ne'
  unfold g
  rw [div_sub_one hν, div_add_one hν, div_div_div_cancel_right₀ hν]
  have h2 : (1 - ν+ ν) = 1 := by ring
  rw [h2, div_one]
  ring

/-- **The census is the gauge without its sign.** -/
theorem census_square (ν : ℝ) : ν* (1 - ν) = (1 - (1 - 2 * ν) ^ 2) / 4 := by ring

/-- The census of a level and of its complement agree. -/
theorem census_complement (ν : ℝ) : ν* (1 - ν) = (1 - ν) * (1 - (1 - ν)) := by ring

/-- **Peel steps add like velocities**, written without dividing by `1 + g₁ g₂`. -/
theorem rapidities_add (z₁ z₂ : ℝ) (h₁ : 0 < z₁) (h₂ : 0 < z₂) :
    g z₁ + g z₂ = g (z₁ * z₂) * (1 + g z₁ * g z₂) := by
  have a : z₁ + 1 ≠0 := by linarith
  have b : z₂ + 1 ≠0 := by linarith
  have c : z₁ * z₂ + 1 ≠0 := by nlinarith
  unfold g
  field_simp
  ring

/-- **Defect split.** The top-left block of `C^2` is `A^2 + BE`. -/
```

```
theorem topleft_sq {m n : Type*} [Fintype m] [Fintype n] [DecidableEq m] [DecidableEq n]
    (A : Matrix m m ℂ) (B : Matrix m n ℂ) (E : Matrix n m ℂ) (D : Matrix n n ℂ) :
    (Matrix.fromBlocks A B E D * Matrix.fromBlocks A B E D).toBlocks₁₁ = A * A + B * E := by
  rw [Matrix.fromBlocks_multiply, Matrix.toBlocks_fromBlocks₁₁]

/-- **Idempotence survives opening.** If `C^2 = C` then `BE = A - A^2`; no Hermiticity is used,
so this holds for the biorthogonal projector of a non-Hermitian band. -/
theorem idempotent_cross {m n : Type*} [Fintype m] [Fintype n] [DecidableEq m] [DecidableEq n]
    (A : Matrix m m ℂ) (B : Matrix m n ℂ) (E : Matrix n m ℂ) (D : Matrix n n ℂ)
    (hC : Matrix.fromBlocks A B E D * Matrix.fromBlocks A B E D = Matrix.fromBlocks A B E D) :
    B * E = A - A * A := by
  have h := congrArg Matrix.toBlocks₁₁ hC
  rw [topleft_sq, Matrix.toBlocks_fromBlocks₁₁] at h
  rw [← h]
  abel

/-- **The exit line.** With `⟨K⟩ = 1 + u^2/p` and `p = E(0)E(π) > 0`, `⟨K⟩ = 2 ↔ p = u^2`. -/
theorem exit_line (u p : ℝ) (hp : 0 < p) : 1 + u ^ 2 / p = 2 ↔ p = u ^ 2 := by
  have hp' : p ≠ 0 := hp.ne'
  constructor
  · intro h
    have : u ^ 2 / p = 1 := by linarith
    rw [div_eq_one_iff_eq hp'] at this
    exact this.symm
  · intro h
    rw [h, div_self (by rw [← h]; exact hp')]
    norm_num

/-- **The nucleus outside the rim.** For a negative first separation energy `-s` and a positive
next one `S > s`, `Γ = (S + s)/(S - s)`. -/
theorem outside_rim (S s : ℝ) (hs : 0 < s) (hS : s < S) :
    g (S / (-s)) = (S + s) / (S - s) := by
  have h1 : (-s) ≠ 0 := by linarith
  have h2 : S - s ≠ 0 := by linarith
  have h3 : S / (-s) + 1 ≠ 0 := by
    rw [div_add_one h1]
    exact div_ne_zero (by linarith) h1
  unfold g
  rw [div_sub_one h1, div_add_one h1, div_div_div_cancel_right₀ h1]
  rw [div_eq_div_iff (by linarith) h2]
  ring

/-- **The rim distance, exactly.** `Γ - 1 = 2s/(S - s)`. -/
theorem rim_distance (S s : ℝ) (hs : 0 < s) (hS : s < S) :
    g (S / (-s)) - 1 = 2 * s / (S - s) := by
  have h2 : S - s ≠ 0 := by linarith
  rw [outside_rim S s hs hS, div_sub_one h2]
  ring_nf

/-- **The Cayley pole.** For `S₁ ≠ 0`, the ratio `S₂/S₁` is `-1` exactly when `S₁ + S₂ = 0`. -/
theorem cayley_pole (S₁ S₂ : ℝ) (h : S₁ ≠ 0) : S₂ / S₁ = -1 ↔ S₁ + S₂ = 0 := by
  rw [div_eq_iff h]
  constructor <;> intro hh <;> linarith

/-- **The ideal gas.** Two Saha ratios share the electron density, which cancels. -/
theorem saha_ratio (n₀ n₁ n₂ x S₀ S₁ : ℝ) (h₀ : n₀ ≠ 0) (h₁ : n₁ ≠ 0) (hx : x ≠ 0)
    (hS : S₀ ≠ 0) (e₀ : n₁ / n₀ = S₀ / x) (e₁ : n₂ / n₁ = S₁ / x) :
    n₀ * n₂ / n₁ ^ 2 = S₁ / S₀ := by
  have a : n₁ = S₀ / x * n₀ := by rw [← e₀]; field_simp
  have b : n₂ = S₁ / x * n₁ := by rw [← e₁]; field_simp
  rw [b, a]
  field_simp
  ring

/-- **The last peel step at first order.** With `I_H = Z^2/2` and `I_He = Z^2/2 - 5Z/8`
(Hartree), the ratio `z = I_H / I_He` has `Γ = 5/(8Z - 5)`. -/
theorem last_step (Z : ℝ) (hZ : 2 ≤ Z) :
    g ((Z ^ 2 / 2) / (Z ^ 2 / 2 - 5 * Z / 8)) = 5 / (8 * Z - 5) := by
```

```
  have h1 : Z ^ 2 / 2 - 5 * Z / 8 ≠0 := by nlinarith
  have h2 : 8 * Z - 5 ≠0 := by linarith
  have h3 : (Z ^ 2 / 2) / (Z ^ 2 / 2 - 5 * Z / 8) + 1 ≠0 := by
    have : 0 < (Z ^ 2 / 2) / (Z ^ 2 / 2 - 5 * Z / 8) := by
      apply div_pos <;> nlinarith
    linarith
  unfold g
  rw [div_eq_div_iff h3 h2]
  field_simp
  ring

end VOTKP.MatterDisk
```

C Scripts and outputs

Every script below is printed whole, with its kill condition in its header, followed by its verbatim output. They run with Python 3, numpy, scipy and matplotlib; the nuclear script also needs pynucastro (for the NUBASE2020 file). Repository: `dicipler-pixel/votkp-lean`, folder `matter-disk/`. The figure scripts write the figures of the text; `s03c` is the raw metric scan that `s03d` repeats with a rank check, kept so the correction can be seen.

`s01_bridge.py`

```
# SCRIPT: MATTER-PEEL-01
# Joins Matter at a Scale (census ladder), VOTKP (Smith/peel coordinate) and the Cone (indefinite NH metric).
import numpy as np
from scipy.integrate import quad
print('='*78); print('MATTER-PEEL-01'); print('='*78)
# (1) identity Gamma = 1-2nu = tanh(eps/2), eps = ln((1-nu)/nu)
nus=np.array([1e-6,0.01,0.2,0.37,0.5,0.73,0.999])
z=(1-nus)/nus; G=(z-1)/(z+1)
print('(1) max|Gamma-(1-2nu)| =',np.max(abs(G-(1-2*nus))),' max|Gamma-tanh(ln z/2)| =',np.max(abs(G-np.tanh(np
    .log(z)/2))))
# (2) entropy over peel rapidity
H=lambda e: np.log(1+np.exp(e))-e*np.exp(e)/(1+np.exp(e)) if e<30 else np.exp(-e)*(1+e)
def Hg(e):
    g=np.tanh(e/2); a,b=(1+g)/2,(1-g)/2
    return -sum(p*np.log(p) for p in (a,b) if p>0)
val,_=quad(Hg,-80,80,limit=500,epsabs=1e-13)
print('(2) int H d(eps) over rapidity =',val,' pi^2/3 =',np.pi**2/3,' diff',abs(val-np.pi**2/3))
```

Output:

```
==============================================================================
MATTER-PEEL-01
==============================================================================
(1) max|Gamma-(1-2nu)| = 1.1102230246251565e-16 max|Gamma-tanh(ln z/2)| = 1.1102230246251565e-16
(2) int H d(eps) over rapidity = 3.2898681336964595 pi^2/3 = 3.289868133696453 diff 6.661338147750939e-15

real 0m0.982s
user 0m0.557s
sys 0m0.121s
```

`s02_ladder.py`

```
# SCRIPT: MATTER-PEEL-02A constant peel step on the measured c=1/2 entanglement ladder
import numpy as np
def binary_H(x):
    x=np.clip(x,1e-15,1-1e-15); return -(x*np.log(x)+(1-x)*np.log(1-x))
```

```
def ladder(h,N,mult=6):
    xi=1/np.log(h); L=int(mult*xi)
    k=2*np.pi*(np.arange(N)+0.5)/N
    e=np.sqrt((h-np.cos(k))**2+np.sin(k)**2); ex=((h-np.cos(k))-1j*np.sin(k))/e
    l=np.arange(-L,L+1); gr=np.array([(ex*np.exp(1j*k*ll)).sum().real/N for ll in l])
    G=np.zeros((2*L,2*L)); ar=np.arange(L)
    for m in range(L):
        G[2*m,1::2]=gr[L-m:L-m+L]; G[2*m+1,0::2]=-gr[L+m-ar]
    nu=np.sort(np.abs(np.linalg.eigvalsh(1j*G)))[::2][::-1]
    p=np.clip((1+nu)/2,0.5,1-1e-15); em=np.sort(np.log(p/(1-p))); em=em[em>1e-10]
    return xi,em
op=lambda u,v:(u+v)/(1+u*v)
for h,N in [(1.05,8192),(1.02,8192),(1.008,16384)]:
    xi,em=ladder(h,N)
    lev=em[::2][:4] # distinct levels (endpoint doublets): eps0,3eps0,5eps0,7eps0
    e0=lev[0]
    unif=np.max(np.abs(lev/e0-np.array([1,3,5,7])))
    Gm=np.tanh(lev/2); step=np.tanh(e0)
    res=[abs(Gm[j+1]-op(Gm[j],step)) for j in range(3)]
    print(f'xi={xi:7.1f} eps0={e0:.4f} Gamma levels={np.round(Gm,5)} peel step tanh(eps0)={step:.5f}')
    print(f' ladder uniformity dev {unif:.4f} | max |Gamma_j+1 - (Gamma_j (+) step)| = {max(res):.2e} (rel to
        step {max(res)/step:.2e})')
```

Output:

```
xi= 20.5 eps0=1.9346 Gamma levels=[0.74751 0.99398 0.99987 1. ] peel step tanh(eps0)=0.95910
   ladder uniformity dev 0.0221 | max |Gamma_j+1 - (Gamma_j (+) step)| = 7.99e-06 (rel to step 8.33e-06)
xi= 50.5 eps0=1.6440 Gamma levels=[0.67615 0.98567 0.99946 0.99998] peel step tanh(eps0)=0.92803
   ladder uniformity dev 0.0113 | max |Gamma_j+1 - (Gamma_j (+) step)| = 5.45e-06 (rel to step 5.87e-06)
xi= 125.5 eps0=1.4275 Gamma levels=[0.61302 0.97276 0.99841 0.99991] peel step tanh(eps0)=0.89115
   ladder uniformity dev 0.0062 | max |Gamma_j+1 - (Gamma_j (+) step)| = 3.23e-06 (rel to step 3.63e-06)

real 0m3.388s
user 0m3.255s
sys 0m0.052s
```

nhssh_lib_and_s03a_occupations.py

```
# SCRIPT: MATTER-PEEL-02B opened matter: NH SSH h(k)=[[iu, v+w e^{-ik}],[v+w e^{ik}, -iu]]
import numpy as np
def bands(v,w,u,k):
    H=np.array([[1j*u, v+w*np.exp(-1j*k)],[v+w*np.exp(1j*k), -1j*u]])
    ev,R=np.linalg.eig(H); o=np.argsort(ev.real); ev,R=ev[o],R[:,o]
    L=np.linalg.inv(R).conj().T # columns: left vectors with <L_m|R_n>=delta
    return ev,R,L
def corr(v,w,u,N,Lc):
    ks=2*np.pi*(np.arange(N)+0.5)/N
    C=np.zeros((2*Lc,2*Lc),complex)
    for k in ks:
        ev,R,L=bands(v,w,u,k)
        P=np.outer(R[:,0],L[:,0].conj()) # occupied lower band, biorthogonal projector
        ph=np.exp(1j*k*np.arange(Lc))
        C+=np.kron(np.outer(ph,ph.conj()),P)
    return C/N
def S_of(nu):
    nu=nu.astype(complex); return -np.sum(nu*np.log(nu)+(1-nu)*np.log(1-nu))
w=1.0
print('B1/B2: biorthogonal occupations -> Gamma=1-2nu, entropy slope')
for (v,u,tag) in [(0.5,0.0,'Hermitian gapped control'),(1.0,0.0,'Hermitian critical control (c=1)'),(0.5,0.3,'
     NH gapped (|v-w|>u)'),(0.5,0.5,'NH critical |v-w|=u (EP)')]:
    N=2000; Ls=[20,40,80,160]; Ss=[]; worst=0; maxim=0
    for Lc in Ls:
        nu=np.linalg.eigvals(corr(v,w,u,N,Lc)); G=1-2*nu
        worst=max(worst,np.max(np.abs(G))); maxim=max(maxim,np.max(np.abs(G.imag)))
        Ss.append(S_of(nu))
```

```
    x=np.log(N/np.pi*np.sin(np.pi*np.array(Ls)/N))
    sl=np.polyfit(x,np.real(Ss),1)[0]
    print(f' {tag:36s} max|Gamma|={worst:9.3e} max|Im Gamma|={maxim:8.2e} c=3*slope={3*sl:+.3f} S(Im)~{np.max(
        np.abs(np.imag(Ss))):.1e}')
```

Output:

```
B1/B2: biorthogonal occupations -> Gamma=1-2nu, entropy slope
  Hermitian gapped control max|Gamma|=1.000e+00 max|Im Gamma|=6.08e-16 c=3*slope=-0.000 S(Im)~9.7e-13
  Hermitian critical control (c=1) max|Gamma|=1.000e+00 max|Im Gamma|=5.69e-16 c=3*slope=+1.000 S(Im)~1.4e-13
  NH gapped (|v-w|>u) max|Gamma|=1.000e+00 max|Im Gamma|=4.20e-01 c=3*slope=+0.000 S(Im)~3.1e-13
  NH critical |v-w|=u (EP) max|Gamma|=3.057e+00 max|Im Gamma|=3.06e+00 c=3*slope=+1.321 S(Im)~2.6e-01

real 0m15.243s
user 0m15.037s
sys 0m0.036s
```

s03b_renyi_central_charge.py

```
# SCRIPT: MATTER-PEEL-02B2 branch-safe Renyi-2 at the NH SSH EP critical point
import numpy as np
exec(open('nhssh_lib_and_s03a_occupations.py').read().split("w=1.0")[0])
w=1.0
def run(v,u,N,Ls):
    S2=[];Sv=[]
    for Lc in Ls:
        nu=np.linalg.eigvals(corr(v,w,u,N,Lc)).astype(complex)
        S2.append(-np.sum(np.log(nu**2+(1-nu)**2)))
    x=np.log(N/np.pi*np.sin(np.pi*np.array(Ls)/N))
    S2=np.array(S2); sl=np.polyfit(x,S2.real,1)[0]
    return 4*sl, np.max(np.abs(S2.imag)), S2.real
for (v,u,tag) in [(1.0,0.0,'Hermitian critical control'),(0.5,0.5,'EP critical, v<w'),(1.5,0.5,'EP critical, v
    >w'),(0.5,0.3,'NH gapped control')]:
    for N in (1000,2000):
        c,im,S=run(v,u,N,[10,20,40,80,160])
        print(f'{tag:28s} N={N}: c from Renyi-2 = {c:+.3f} max|Im S2|={im:.1e} S2={np.round(S,3)}')
```

Output:

```
Hermitian critical control N=1000: c from Renyi-2 = +1.007 max|Im S2|=7.7e-15 S2=[1.321 1.497 1.671 1.843
    2.008]
Hermitian critical control N=2000: c from Renyi-2 = +1.007 max|Im S2|=2.2e-15 S2=[1.321 1.497 1.671 1.845
    2.016]
EP critical, v<w N=1000: c from Renyi-2 = -1.792 max|Im S2|=6.3e+00 S2=[-1.942 -2.351 -2.69 -2.963 -3.166]
EP critical, v<w N=2000: c from Renyi-2 = -1.956 max|Im S2|=6.3e+00 S2=[-2.209 -2.633 -2.997 -3.307 -3.56 ]
EP critical, v>w N=1000: c from Renyi-2 = -0.876 max|Im S2|=3.7e-14 S2=[-0.901 -1.074 -1.238 -1.383 -1.495]
EP critical, v>w N=2000: c from Renyi-2 = -1.089 max|Im S2|=1.1e-13 S2=[-1.055 -1.257 -1.455 -1.641 -1.803]
NH gapped control N=1000: c from Renyi-2 = +0.000 max|Im S2|=2.7e-15 S2=[1.778 1.778 1.778 1.778 1.778]
NH gapped control N=2000: c from Renyi-2 = +0.000 max|Im S2|=1.6e-14 S2=[1.778 1.778 1.778 1.778 1.778]

real 0m26.878s
user 0m26.613s
sys 0m0.024s
```

s03c_metric_raw.py

```
# SCRIPT: MATTER-PEEL-02B3 signature of the round-trip form on the (k, v) plane, NH SSH
import numpy as np
w=1.0
def Hm(v,u,k): return np.array([[1j*u, v+w*np.exp(-1j*k)],[v+w*np.exp(1j*k), -1j*u]])
def form(v,u,k):
    H=Hm(v,u,k); ev,R=np.linalg.eig(H); o=np.argsort(ev.real); ev,R=ev[o],R[:,o]
    L=np.linalg.inv(R).conj().T
```

```
    dk=np.array([[0,-1j*w*np.exp(-1j*k)],[1j*w*np.exp(1j*k),0]]); dv=np.array([[0,1],[1,0]],complex)
    def Q(V): return (L[:,1].conj()@V@R[:,0])*(L[:,0].conj()@V@R[:,1])/(ev[1]-ev[0])**2
    a=Q(dk).real; c=Q(dv).real; b=Q((dk+dv)/np.sqrt(2)).real-(a+c)/2
    return a,b,c
def scan(u):
    ks=np.linspace(0.02,2*np.pi-0.02,180); vs=np.linspace(0.05,1.95,96)
    lor=np.zeros((len(vs),len(ks)),bool); defin=np.zeros_like(lor)
    for i,v in enumerate(vs):
        for j,k in enumerate(ks):
            if abs(v+w*np.exp(1j*k))**2-u**2 < 1e-6: continue # skip exceptional points themselves
            a,b,c=form(v,u,k); d=a*c-b*b
            lor[i,j]=d<0; defin[i,j]=d>0
    return ks,vs,lor,defin
for u in (0.0,0.5):
    ks,vs,lor,defin=scan(u)
    print(f'u={u}: Lorentzian fraction {lor.mean():.3f} definite fraction {defin.mean():.3f}')
    if lor.any():
        idx=np.argwhere(lor); kk=ks[idx[:,1]]; vv=vs[idx[:,0]]
        dist=np.min(np.hypot(kk-np.pi, vv-0.5))
        print(f' Lorentzian cells span k in [{kk.min():.2f},{kk.max():.2f}], v in [{vv.min():.2f},{vv.max():.2f
            }]; nearest to EP (pi,0.5): {dist:.3f}')
        # PT-broken region |v+w e^{ik}|<u
        brk=np.array([[abs(v+w*np.exp(1j*k))<u for k in ks] for v in vs])
        print(f' Lorentzian cells inside PT-broken zone: {np.mean(brk[lor]):.3f}; PT-broken cells that are
            Lorentzian: {np.mean(lor[brk]) if brk.any() else float("nan"):.3f}')
        np.save('lor_u05.npy',lor); np.save('brk_u05.npy',brk)
```

Output:

```
u=0.0: Lorentzian fraction 0.338 definite fraction 0.525
   Lorentzian cells span k in [0.02,6.26], v in [0.05,1.95]; nearest to EP (pi,0.5): 0.020
   Lorentzian cells inside PT-broken zone: 0.000; PT-broken cells that are Lorentzian: nan
u=0.5: Lorentzian fraction 0.932 definite fraction 0.000
   Lorentzian cells span k in [0.02,6.26], v in [0.05,1.95]; nearest to EP (pi,0.5): 0.020
   Lorentzian cells inside PT-broken zone: 0.000; PT-broken cells that are Lorentzian: 0.000

real 0m2.668s
user 0m2.656s
sys 0m0.008s
```

s03d_metric_rank_checked.py

```
# SCRIPT: MATTER-PEEL-02B3b signature map with the rank pre-check (degenerate-plane artifact gated)
import numpy as np
exec(open('s03c_metric_raw.py').read().split('def scan')[0])
def classify(v,u,k,tol=1e-9):
    a,b,c=form(v,u,k); d=a*c-b*b; s=a*a+c*c+2*b*b
    if abs(d)<=tol*s: return 0 # rank-one / degenerate: no verdict
    return -1 if d<0 else 1
ks=np.linspace(0.02,2*np.pi-0.02,180); vs=np.linspace(0.05,1.95,96)
for u in (0.0,0.5):
    M=np.array([[classify(v,u,k) if abs(abs(v+w*np.exp(1j*k))**2-u**2)>1e-6 else 0 for k in ks] for v in vs])
    brk=np.array([[abs(v+w*np.exp(1j*k))<u for k in ks] for v in vs])
    print(f'u={u}: Lorentzian {np.mean(M==-1):.3f} definite {np.mean(M==1):.3f} degenerate {np.mean(M==0):.3f
        }')
    if (M==-1).any():
        idx=np.argwhere(M==-1); kk=ks[idx[:,1]]; vv=vs[idx[:,0]]
        print(f' Lorentzian: k in [{kk.min():.2f},{kk.max():.2f}] v in [{vv.min():.2f},{vv.max():.2f}]; nearest
            to EP {np.min(np.hypot(kk-np.pi,vv-0.5)):.3f}')
        print(f' fraction of Lorentzian cells in PT-broken zone {np.mean(brk[M==-1]):.3f}; PT-broken cells
            Lorentzian {np.mean((M==-1)[brk]) if brk.any() else float("nan"):.3f}')
    np.save(f'sig_u{u}.npy',M)
# radial look at the EP: signature along k=pi as v crosses 0.5, u=0.5
print('along k=pi:')
for v in [0.2,0.4,0.45,0.49,0.51,0.55,0.6,0.8,1.2,1.49,1.51,1.8]:
    a,b,c=form(v,0.5,np.pi); print(f' v={v:4.2f} a={a:+.3e} b={b:+.3e} c={c:+.3e} det={a*c-b*b:+.3e}')
```

Output:

```
u=0.0: Lorentzian 0.000 definite 0.000 degenerate 1.000
u=0.5: Lorentzian 0.932 definite 0.068 degenerate 0.000
   Lorentzian: k in [0.02,6.26] v in [0.05,1.95]; nearest to EP 0.020
   fraction of Lorentzian cells in PT-broken zone 0.000; PT-broken cells Lorentzian 0.000
along k=pi:
  v=0.20 a=+6.410e-01 b=-3.608e-16 c=-4.109e-01 det=-2.634e-01
  v=0.40 a=+2.273e+00 b=-2.220e-16 c=-5.165e+00 det=-1.174e+01
  v=0.45 a=+4.762e+00 b=+3.553e-15 c=-2.268e+01 det=-1.080e+02
  v=0.49 a=+2.475e+01 b=-5.684e-14 c=-6.127e+02 det=-1.517e+04
  v=0.51 a=-2.525e+01 b=-5.684e-14 c=-6.377e+02 det=+1.610e+04
  v=0.55 a=-5.263e+00 b=+7.105e-15 c=-2.770e+01 det=+1.458e+02
  v=0.60 a=-2.778e+00 b=+0.000e+00 c=-7.716e+00 det=+2.143e+01
  v=0.80 a=-1.190e+00 b=+2.220e-16 c=-1.417e+00 det=+1.687e+00
  v=1.20 a=-1.190e+00 b=+4.441e-16 c=-1.417e+00 det=+1.687e+00
  v=1.49 a=-2.525e+01 b=+2.842e-13 c=-6.377e+02 det=+1.610e+04
  v=1.51 a=+2.475e+01 b=+2.274e-13 c=-6.127e+02 det=-1.517e+04
  v=1.80 a=+6.410e-01 b=+2.776e-17 c=-4.109e-01 det=-2.634e-01

real 0m3.778s
user 0m3.739s
sys 0m0.016s
```

s04a_rotation_unwindowed.py

```
# SCRIPT: MATTER-PEEL-03 rapidity across the Cayley pole: rotation part of census levels on approach to the
     critical wall
import numpy as np
exec(open('nhssh_lib_and_s03a_occupations.py').read().split('w=1.0')[0]); w=1.0
# exact algebra check: |tanh(W/2)|<1 iff |Im W|<pi/2 (principal branch)
rng=np.random.default_rng(1); Z=rng.normal(size=4000)+1j*rng.normal(size=4000); W=np.log(Z)
print('theorem check, mismatches:',np.sum((np.abs(np.tanh(W/2))<1)!=(np.abs(W.imag)<np.pi/2)))
for zN in [-28.0]:
    print(' negative ratio z=',zN,' Im W=',np.log(complex(zN)).imag/np.pi,'pi Gamma=',np.tanh(np.log(complex(
        zN))/2))
def census_rot(v,u,N=2000,Lc=80):
    nu=np.linalg.eigvals(corr(v,w,u,N,Lc)).astype(complex)
    z=(1-nu)/nu; W=np.log(z); G=1-2*nu
    return np.max(np.abs(W.imag))/np.pi, np.max(np.abs(G))
u=0.5
print('approach from v<w side (critical at v=0.5): v, max|Im W|/pi, max|Gamma|')
for v in [0.0,0.1,0.2,0.3,0.4,0.45,0.48,0.49,0.495,0.5]:
    r,g=census_rot(v,u); print(f' v={v:5.3f} rot={r:.4f} pi |Gamma|max={g:.4f} {"OUT" if g>1+1e-9 else "in"}')
print('opposite: approach from v>w side (critical at v=1.5)')
for v in [2.5,2.0,1.7,1.6,1.55,1.52,1.51,1.505,1.5]:
    r,g=census_rot(v,u); print(f' v={v:5.3f} rot={r:.4f} pi |Gamma|max={g:.4f} {"OUT" if g>1+1e-9 else "in"}')
print('inside the PT-broken zone (0.5<v<1.5):')
for v in [0.6,1.0,1.4]:
    r,g=census_rot(v,u); print(f' v={v:5.3f} rot={r:.4f} pi |Gamma|max={g:.4f} {"OUT" if g>1+1e-9 else "in"}')
```

Output:

```
theorem check, mismatches: 0
 negative ratio z= -28.0 Im W= 1.0 pi Gamma= (1.0740740740740737+9.407437688923424e-18j)
approach from v<w side (critical at v=0.5): v, max|Im W|/pi, max|Gamma|
  v=0.000 rot=0.9998 pi |Gamma|max=1.0000 in
  v=0.100 rot=0.9993 pi |Gamma|max=1.0000 in
  v=0.200 rot=0.9981 pi |Gamma|max=1.0000 in
  v=0.300 rot=1.0000 pi |Gamma|max=1.0000 in
  v=0.400 rot=0.9995 pi |Gamma|max=1.0000 in
  v=0.450 rot=0.9997 pi |Gamma|max=1.0000 in
  v=0.480 rot=1.0000 pi |Gamma|max=1.0927 OUT
  v=0.490 rot=0.9978 pi |Gamma|max=1.2326 OUT
  v=0.495 rot=0.9954 pi |Gamma|max=1.3724 OUT
  v=0.500 rot=1.0000 pi |Gamma|max=2.9595 OUT
```

```
opposite: approach from v>w side (critical at v=1.5)
  v=2.500 rot=1.0000 pi |Gamma|max=1.0000 in
  v=2.000 rot=0.9994 pi |Gamma|max=1.0000 in
  v=1.700 rot=1.0000 pi |Gamma|max=1.0000 in
  v=1.600 rot=0.9998 pi |Gamma|max=1.0000 in
  v=1.550 rot=0.9990 pi |Gamma|max=1.0000 in
  v=1.520 rot=0.9993 pi |Gamma|max=1.0000 in
  v=1.510 rot=1.0000 pi |Gamma|max=1.0009 OUT
  v=1.505 rot=1.0000 pi |Gamma|max=1.0301 OUT
  v=1.500 rot=1.0000 pi |Gamma|max=1.4681 OUT
inside the PT-broken zone (0.5<v<1.5):
  v=0.600 rot=0.9995 pi |Gamma|max=1.0000 in
  v=1.000 rot=0.9982 pi |Gamma|max=1.0000 in
  v=1.400 rot=0.9992 pi |Gamma|max=1.0000 in

real 0m17.028s
user 0m16.805s
sys 0m0.020s
```

s04b_rotation_windowed.py

```
# SCRIPT: MATTER-PEEL-03b windowed census rotation and disk exit vs block size
import numpy as np
exec(open('nhssh_lib_and_s03a_occupations.py').read().split('w=1.0')[0]); w=1.0
def census(v,u,N,Lc,d=1e-9):
    nu=np.linalg.eigvals(corr(v,w,u,N,Lc)).astype(complex)
    keep=(np.abs(nu)>d)&(np.abs(1-nu)>d); nu=nu[keep]
    W=np.log((1-nu)/nu); G=1-2*nu
    return (np.max(np.abs(W.imag))/np.pi if len(nu) else 0.0), np.max(np.abs(G)) if len(nu) else 0.0
u=0.5
for (N,Lc) in [(2000,40),(2000,80),(4000,160)]:
    print(f'block Lc={Lc}, ring N={N}')
    for v in [0.3,0.4,0.45,0.47,0.48,0.49,0.495,0.5,1.505,1.51,1.52,1.55]:
        r,g=census(v,u,N,Lc)
        print(f' v={v:5.3f} max rotation={r:.4f} pi max|Gamma|={g:.4f} {"OUT" if g>1+1e-6 else "in"}')
```

Output:

```
block Lc=40, ring N=2000
   v=0.300 max rotation=0.3734 pi max|Gamma|=1.0000 in
   v=0.400 max rotation=0.4218 pi max|Gamma|=1.0000 in
   v=0.450 max rotation=0.4703 pi max|Gamma|=1.0000 in
   v=0.470 max rotation=0.5036 pi max|Gamma|=1.0115 OUT
   v=0.480 max rotation=0.5282 pi max|Gamma|=1.0927 OUT
   v=0.490 max rotation=0.5686 pi max|Gamma|=1.2326 OUT
   v=0.495 max rotation=0.6105 pi max|Gamma|=1.3724 OUT
   v=0.500 max rotation=1.0000 pi max|Gamma|=2.8380 OUT
   v=1.505 max rotation=0.5468 pi max|Gamma|=1.0301 OUT
   v=1.510 max rotation=0.5014 pi max|Gamma|=1.0008 OUT
   v=1.520 max rotation=0.4616 pi max|Gamma|=1.0000 in
   v=1.550 max rotation=0.4005 pi max|Gamma|=1.0000 in
block Lc=80, ring N=2000
   v=0.300 max rotation=0.3734 pi max|Gamma|=1.0000 in
   v=0.400 max rotation=0.4218 pi max|Gamma|=1.0000 in
   v=0.450 max rotation=0.4703 pi max|Gamma|=1.0000 in
   v=0.470 max rotation=0.5036 pi max|Gamma|=1.0115 OUT
   v=0.480 max rotation=0.5282 pi max|Gamma|=1.0927 OUT
   v=0.490 max rotation=0.5661 pi max|Gamma|=1.2326 OUT
   v=0.495 max rotation=0.5992 pi max|Gamma|=1.3724 OUT
   v=0.500 max rotation=1.0000 pi max|Gamma|=2.9595 OUT
   v=1.505 max rotation=0.5376 pi max|Gamma|=1.0301 OUT
   v=1.510 max rotation=0.5012 pi max|Gamma|=1.0009 OUT
   v=1.520 max rotation=0.4616 pi max|Gamma|=1.0000 in
   v=1.550 max rotation=0.4005 pi max|Gamma|=1.0000 in
block Lc=160, ring N=4000
```

```
   v=0.300 max rotation=0.3734 pi max|Gamma|=1.0000 in
   v=0.400 max rotation=0.4218 pi max|Gamma|=1.0000 in
   v=0.450 max rotation=0.4703 pi max|Gamma|=1.0000 in
   v=0.470 max rotation=0.5036 pi max|Gamma|=1.0115 OUT
   v=0.480 max rotation=0.5282 pi max|Gamma|=1.0927 OUT
   v=0.490 max rotation=0.5661 pi max|Gamma|=1.2326 OUT
   v=0.495 max rotation=0.5991 pi max|Gamma|=1.3724 OUT
   v=0.500 max rotation=1.0000 pi max|Gamma|=3.2318 OUT
   v=1.505 max rotation=0.5357 pi max|Gamma|=1.0301 OUT
   v=1.510 max rotation=0.5012 pi max|Gamma|=1.0009 OUT
   v=1.520 max rotation=0.4616 pi max|Gamma|=1.0000 in
   v=1.550 max rotation=0.4005 pi max|Gamma|=1.0000 in

real 1m21.279s
user 1m20.663s
sys 0m0.049s
```

s05a_exit_line.py

```
# SCRIPT: MATTER-PEEL-03c locate the disk-exit line v_exit(u) and the rotation at the critical point
import numpy as np
exec(open('nhssh_lib_and_s03a_occupations.py').read().split('w=1.0')[0]); w=1.0
def census(v,u,N=2000,Lc=40,d=1e-9):
    nu=np.linalg.eigvals(corr(v,w,u,N,Lc)).astype(complex)
    keep=(np.abs(nu)>d)&(np.abs(1-nu)>d); nu=nu[keep]
    W=np.log((1-nu)/nu); return np.max(np.abs(W.imag))/np.pi
def vexit(u):
    lo,hi=0.0,1-u # rotation <1/2 at v=0 (check), critical at v=1-u
    if census(lo,u)>=0.5: return None
    for _ in range(30):
        m=(lo+hi)/2
        if census(m,u)<0.5: lo=m
        else: hi=m
    return (lo+hi)/2
print(' u v_exit crit=1-u gap at exit |v-w|-u ratio u/(w-v_exit) rot at critical')
for u in [0.2,0.3,0.4,0.5,0.6,0.7]:
    ve=vexit(u); rc=census(1-u,u)
    if ve is None: print(f' {u:.2f} rotation already >= 1/2 at v=0; rot at critical {rc:.4f}'); continue
    print(f' {u:.2f} {ve:.6f} {1-u:.3f} {abs(ve-w)-u:.6f} {u/(w-ve):.6f} {rc:.5f} pi')
```

Output:

```
 u v_exit crit=1-u gap at exit |v-w|-u ratio u/(w-v_exit) rot at critical
 0.20 0.798752 0.800 0.001248 0.993801 1.00000 pi
 0.30 0.695189 0.700 0.004811 0.984216 1.00000 pi
 0.40 0.586649 0.600 0.013351 0.967700 1.00000 pi
 0.50 0.468213 0.500 0.031787 0.940226 1.00000 pi
 0.60 0.327358 0.400 0.072642 0.892005 1.00000 pi
 0.70 0.082016 0.300 0.217984 0.762540 1.00000 pi

real 1m11.323s
user 1m10.492s
sys 0m0.024s
```

s05b_predict_exit.py

```
# SCRIPT: MATTER-PEEL-03d predict the disk-exit line from mean Petermann = 2, then measure
import numpy as np
exec(open('s05a_exit_line.py').read().split("print(' u ")[0])
def P(v,u,k):
    H=np.array([[1j*u, v+w*np.exp(-1j*k)],[v+w*np.exp(1j*k), -1j*u]])
    ev,R=np.linalg.eig(H); o=np.argsort(ev.real); R=R[:,o]; L=np.linalg.inv(R).conj().T
    return np.outer(R[:,0],L[:,0].conj())
```

```
ks=2*np.pi*(np.arange(4000)+0.5)/4000
def meanK(v,u): return np.mean([np.linalg.norm(P(v,u,k),2)**2 for k in ks])
def vpred(u):
    lo,hi=0.0,1-u-1e-9
    for _ in range(40):
        m=(lo+hi)/2
        if meanK(m,u)<2: lo=m
        else: hi=m
    return (lo+hi)/2
print(' u predicted (mean K = 2) measured census exit difference')
for u in [0.25,0.45,0.65]:
    vp=vpred(u); vm=vexit(u)
    print(f' {u:.2f} {vp:.7f} {vm:.7f} {vm-vp:+.2e}')
```

Output:

```
 u predicted (mean K = 2) measured census exit difference
 0.25 0.7474014 0.7474015 +4.04e-08
 0.45 0.5291503 0.5291503 -1.53e-09
 0.65 0.2357262 0.2357262 -3.70e-08

real 1m14.012s
user 1m13.157s
sys 0m0.042s
```

s06_second_model_exit_law.py

```
# SCRIPT: MATTER-DISK-S06 is the disk-exit law universal? second model: the cone paper's band in 1D
# WHY THIS CHECK PROVES WHAT IT CLAIMS
# Section 5 found, on the PT-symmetric SSH chain, that the census levels leave the Smith disk exactly
# where the band-averaged Petermann factor <K> equals 2. For any two-band h(k) = d(k).sigma the
# Petermann factor of the lower band is K(k) = (1 + |d|^2/|E|^2)/2 with E^2 = d.d (checked below).
# A law that belongs to the SSH chain alone would fail on a different band. Here the band is the cone
# paper's family made one-dimensional, d = (t1 + cos k, sin k, m + i*gamma): a mass term, no PT symmetry.
# Path: start closed (gamma = 0, census real, inside the disk) and raise gamma until the census leaves
# the disk (largest rotation |Im ln z| reaches pi/2, census window 1e-9 < nu < 1-1e-9).
# KILL (written before the run): at any of the four (t1, m) settings, <K> at the measured exit differs
# from 2 by more than 1e-4. Control: gamma = 0 must give a real census inside the disk.
import numpy as np
sx=np.array([[0,1],[1,0]],complex); sy=np.array([[0,-1j],[1j,0]]); sz=np.array([[1,0],[0,-1]],complex)
def H(t1,m,g,k): return (t1+np.cos(k))*sx+np.sin(k)*sy+(m+1j*g)*sz
def lower(t1,m,g,k):
    ev,R=np.linalg.eig(H(t1,m,g,k)); o=np.argsort(ev.real); ev,R=ev[o],R[:,o]
    L=np.linalg.inv(R).conj().T; return ev,R,L
def corr(t1,m,g,N,Lc):
    ks=2*np.pi*(np.arange(N)+0.5)/N; C=np.zeros((2*Lc,2*Lc),complex)
    for k in ks:
        ev,R,L=lower(t1,m,g,k); P=np.outer(R[:,0],L[:,0].conj())
        ph=np.exp(1j*k*np.arange(Lc)); C+=np.kron(np.outer(ph,ph.conj()),P)
    return C/N
def rotation(t1,m,g,N=2000,Lc=40,d=1e-9):
    nu=np.linalg.eigvals(corr(t1,m,g,N,Lc)).astype(complex)
    keep=(np.abs(nu)>d)&(np.abs(1-nu)>d); nu=nu[keep]
    if len(nu)==0: return 0.0, 0.0
    W=np.log((1-nu)/nu); return np.max(np.abs(W.imag))/np.pi, np.max(np.abs(1-2*nu))
ks=2*np.pi*(np.arange(4000)+0.5)/4000
def meanK(t1,m,g):
    out=[]
    for k in ks:
        ev,R,L=lower(t1,m,g,k); out.append(np.vdot(L[:,0],L[:,0]).real*np.vdot(R[:,0],R[:,0]).real)
    return np.mean(out)
# check the general Petermann formula on random points
rng=np.random.default_rng(5); worst=0
for _ in range(300):
    t1,m,g,k=rng.uniform(0.2,2),rng.uniform(-1,1),rng.uniform(0,1.5),rng.uniform(0,2*np.pi)
```

```
    ev,R,L=lower(t1,m,g,k); d=np.array([t1+np.cos(k),np.sin(k),m+1j*g]); E2=np.sum(d*d)
    if abs(E2)<1e-3: continue
    Kn=np.vdot(L[:,0],L[:,0]).real*np.vdot(R[:,0],R[:,0]).real; Kf=(1+np.sum(abs(d)**2)/abs(E2))/2
    worst=max(worst,abs(Kn-Kf))
print('general Petermann formula K=(1+|d|^2/|E|^2)/2, worst error on 300 random points:',worst)
print('settings (t1, m): control at gamma=0, measured exit gamma, <K> at exit')
ok=True
for t1,m in [(0.5,0.4),(1.5,0.4),(0.5,0.0),(1.5,0.8)]:
    r0,G0=rotation(t1,m,0.0)
    lo,hi=0.0,None; g=0.05
    while hi is None and g<4:
        if rotation(t1,m,g)[0]>=0.5: hi=g
        else: lo=g; g*=1.4
    if hi is None: print(f' ({t1},{m}): census never left the disk up to gamma=4'); ok=False; continue
    for _ in range(28):
        mid=(lo+hi)/2
        if rotation(t1,m,mid)[0]<0.5: lo=mid
        else: hi=mid
    ge=(lo+hi)/2; K=meanK(t1,m,ge)
    ok&=abs(K-2)<1e-4
    print(f' ({t1},{m}): control rot={r0:.3f} pi max|Gamma|={G0:.4f} | exit gamma={ge:.7f} | <K>={K:.6f} {"
        PASS" if abs(K-2)<1e-4 else "FAIL"}')
print('VERDICT:', 'exit law <K> = 2 holds on the second model' if ok else 'KILLED on the second model at the
    stated bar')
```

Output:

```
general Petermann formula K=(1+|d|^2/|E|^2)/2, worst error on 300 random points: 1.4921397450962104e-13
settings (t1, m): control at gamma=0, measured exit gamma, <K> at exit
  (0.5,0.4): control rot=0.000 pi max|Gamma|=1.0000 | exit gamma=0.1691699 | <K>=1.022352 FAIL
  (1.5,0.4): control rot=0.000 pi max|Gamma|=1.0000 | exit gamma=0.1956728 | <K>=1.018838 FAIL
  (0.5,0.0): control rot=0.000 pi max|Gamma|=1.0000 | exit gamma=0.4743416 | <K>=2.000000 PASS
  (1.5,0.8): control rot=0.000 pi max|Gamma|=1.0000 | exit gamma=0.2518871 | <K>=1.015041 FAIL
VERDICT: KILLED on the second model at the stated bar

real 0m57.253s
user 0m56.558s
sys 0m0.044s
block 40, ring 2000: exit gamma = 0.169170
block 80, ring 2000: exit gamma = 0.169170
block 160, ring 4000: exit gamma = 0.169170
```

s07_boundary_census_disk.py

```
# SCRIPT: MATTER-DISK-S07 the boundary census on the Smith disk, closed, warm and opened
# WHY THIS CHECK PROVES WHAT IT CLAIMS
# For a one-body correlation matrix C split into a region A and the rest B, block multiplication gives
# C_A - C_A^2 = (C - C^2)_A + X_A, X_A = C_AB C_BA.
# In a pure state C^2 = C, so X_A = C_A(1 - C_A): its eigenvalues are nu(1 - nu) = (1 - Gamma^2)/4 with
# Gamma = 1 - 2nu, the section-1 disk coordinate. So the boundary coordinate Gamma_b = sqrt(1 - 4 lambda(X_A))
# is the same point on the disk as |Gamma|, and the count lambda >= eps(1-eps) is the disk window
# |Gamma_b| <= 1 - 2eps. In a warm state C^2 != C: the local mixing term (C - C^2)_A pulls raw Gamma toward
# the centre without any correlation across the cut; X_A keeps only what crosses the cut.
# KILL (written before the run):
# (a) the identity fails by more than 1e-12 on 32 random Hermitian contractions;
# (b) disconnected sites C = 0.3 I give a nonzero boundary count;
# (c) at T = 0 the boundary count differs from the raw count, or |Gamma_b| differs from |Gamma| by > 1e-8;
# (d) the 128-site ring table (region 64, eps = 0.1) differs from the 3 Nov review table by more than
# 1e-5 in Tr X_A or mutual information, or by any integer in the counts;
# (e) on the opened non-Hermitian SSH chain (biorthogonal C) C^2 = C or the identity fails by > 1e-8.
import numpy as np
from numpy.linalg import eigvalsh, eigvals, eigh
rng=np.random.default_rng(7)
def blocks(C,nA): return C[:nA,:nA],C[:nA,nA:],C[nA:,:nA]
def ident_err(C,nA):
```

```
    CA,CAB,CBA=blocks(C,nA); CC=C@C
    return np.max(np.abs((CA-CA@CA)-((C-CC)[:nA,:nA]+CAB@CBA)))
# (a) random Hermitian contractions 0 <= C <= I
worst=0
for s in range(32):
    n=24; M=rng.normal(size=(n,n))+1j*rng.normal(size=(n,n)); U,_=np.linalg.qr(M)
    C=U@np.diag(rng.uniform(0,1,n))@U.conj().T; worst=max(worst,ident_err(C,10))
print('(a) block identity, worst error on 32 random Hermitian contractions:',f'{worst:.2e}', 'PASS' if worst<1
     e-12 else 'FAIL')
eps=0.1; thr=eps*(1-eps)
def counts(C,nA):
    CA,CAB,CBA=blocks(C,nA); nu=eigvalsh(CA); X=CAB@CBA; lam=eigvalsh((X+X.conj().T)/2)
    raw=int(np.sum((nu>=eps)&(nu<=1-eps))); bnd=int(np.sum(lam>=thr))
    return raw,bnd,np.trace(X).real,nu,lam
# (b) disconnected sites
C=0.3*np.eye(64); raw,bnd,tx,_,_=counts(C,32)
print(f'(b) disconnected C=0.3 I, region 32: raw count {raw}, boundary count {bnd}, Tr X_A {tx:.3f}',
      'PASS' if bnd==0 else 'FAIL')
# (c,d) the 128-site antiperiodic ring
def ring(tA,tB,N=128):
    H=np.zeros((N,N))
    for j in range(N):
        t=tA if j%2==0 else tB; s=-1.0 if j==N-1 else 1.0
        H[j,(j+1)%N]=H[(j+1)%N,j]=-t*s
    return H
def thermal(H,T):
    e,V=eigh(H)
    f=(e<0).astype(float) if T==0 else 1/(1+np.exp(e/T))
    return (V*f)@V.T
def Sg(nu):
    nu=np.clip(nu,1e-15,1-1e-15); return -np.sum(nu*np.log(nu)+(1-nu)*np.log(1-nu))
review={('uniform',0):(2,2,0.605657,3.923714),('uniform',0.5):(24,0,0.170835,0.949562),('uniform',2)
     :(64,0,0.027779,0.116547),
       ('gapped alternating',0):(0,0,0.029017,0.345837),('gapped alternating',0.5):(10,0,0.024935,0.188891),('
            gapped alternating',2):(64,0,0.006484,0.028357)}
okd=True; okc=True
print('(d) ring N=128 antiperiodic, mu=0, region 64, eps=0.1 [ours | 3 Nov review]')
for name,(tA,tB) in [('uniform',(1,1)),('gapped alternating',(1.5,0.5))]:
    H=ring(tA,tB)
    for T in [0,0.5,2]:
        C=thermal(H,T); raw,bnd,tx,nu,lam=counts(C,64)
        MI=Sg(eigvalsh(C[:64,:64]))+Sg(eigvalsh(C[64:,64:]))-Sg(eigvalsh(C))
        r=review[(name,T)]
        good=(raw==r[0])and(bnd==r[1])and abs(tx-r[2])<1e-5 and abs(MI-r[3])<1e-5; okd&=good
        print(f' {name:19s} T={T:<3} raw {raw:3d}|{r[0]:3d} boundary {bnd:2d}|{r[1]:2d} TrX {tx:.6f}|{r[2]:.6f}
              MI {MI:.6f}|{r[3]:.6f} {"MATCH" if good else "DIFFER"}')
        G=np.sort(np.abs(1-2*nu)); Gb=np.sort(np.sqrt(np.clip(1-4*lam,0,1)))
        if T==0:
            dg=np.max(np.abs(G-Gb)); okc&=(raw==bnd) and dg<1e-8
            print(f' (c) T=0 disk check: max | |Gamma| - |Gamma_b| | = {dg:.1e}')
        else:
            print(f' warm: levels inside the window by raw Gamma {raw}, by boundary Gamma_b {bnd}; innermost |
                 Gamma| {G[0]:.3f}, innermost |Gamma_b| {Gb[0]:.3f}')
print('(c) T=0 boundary census = original census on the disk:', 'PASS' if okc else 'FAIL')
print('(d) review table reproduced:', 'PASS' if okd else 'DIFFER (see rows)')
# (e) opened matter: biorthogonal C of the non-Hermitian SSH chain
def bands(v,w,u,k):
    Hk=np.array([[1j*u, v+w*np.exp(-1j*k)],[v+w*np.exp(1j*k), -1j*u]])
    ev,R=np.linalg.eig(Hk); o=np.argsort(ev.real); ev,R=ev[o],R[:,o]; L=np.linalg.inv(R).conj().T
    return ev,R,L
def corr(v,w,u,N,Lc):
    ks=2*np.pi*(np.arange(N)+0.5)/N; C=np.zeros((2*Lc,2*Lc),complex)
    for k in ks:
        ev,R,L=bands(v,w,u,k); P=np.outer(R[:,0],L[:,0].conj()); ph=np.exp(1j*k*np.arange(Lc))
        C+=np.kron(np.outer(ph,ph.conj()),P)
    return C/N
oke=True
print('(e) opened chain, full ring of 200 cells (C is the whole system), block of 40 cells')
```

```
for v,u,tag in [(0.3,0.5,'opened, inside the disk'),(0.5,0.5,'opened, critical')]:
    C=corr(v,1.0,u,200,200); idem=np.max(np.abs(C@C-C)); ie=ident_err(C,80)
    CA,CAB,CBA=blocks(C,80); pure=np.max(np.abs(CAB@CBA-(CA-CA@CA)))
    oke&=idem<1e-8 and ie<1e-8 and pure<1e-8
    print(f' v={v} u={u} ({tag}): |C^2 - C| = {idem:.1e}, identity error {ie:.1e}, |X_A - C_A(1-C_A)| = {pure
        :.1e}')
print('(e) opened matter keeps the boundary identity:', 'PASS' if oke else 'FAIL')
```

Output:

```
(a) block identity, worst error on 32 random Hermitian contractions: 2.78e-16 PASS
(b) disconnected C=0.3 I, region 32: raw count 32, boundary count 0, Tr X_A 0.000 PASS
(d) ring N=128 antiperiodic, mu=0, region 64, eps=0.1 [ours | 3 Nov review]
    uniform T=0 raw 2| 2 boundary 2| 2 TrX 0.605657|0.605657 MI 3.923714|3.923714 MATCH
        (c) T=0 disk check: max | |Gamma| - |Gamma_b| | = 4.0e-15
    uniform T=0.5 raw 24| 24 boundary 0| 0 TrX 0.170835|0.170835 MI 0.949562|0.949562 MATCH
        warm: levels inside the window by raw Gamma 24, by boundary Gamma_b 0; innermost |Gamma| 0.047,
            innermost |Gamma_b| 0.817
    uniform T=2 raw 64| 64 boundary 0| 0 TrX 0.027779|0.027779 MI 0.116547|0.116547 MATCH
        warm: levels inside the window by raw Gamma 64, by boundary Gamma_b 0; innermost |Gamma| 0.012,
            innermost |Gamma_b| 0.972
    gapped alternating T=0 raw 0| 0 boundary 0| 0 TrX 0.029017|0.029017 MI 0.345837|0.345837 MATCH
        (c) T=0 disk check: max | |Gamma| - |Gamma_b| | = 4.2e-15
    gapped alternating T=0.5 raw 10| 10 boundary 0| 0 TrX 0.024935|0.024935 MI 0.188891|0.188891 MATCH
        warm: levels inside the window by raw Gamma 10, by boundary Gamma_b 0; innermost |Gamma| 0.763,
            innermost |Gamma_b| 0.980
    gapped alternating T=2 raw 64| 64 boundary 0| 0 TrX 0.006484|0.006484 MI 0.028357|0.028357 MATCH
        warm: levels inside the window by raw Gamma 64, by boundary Gamma_b 0; innermost |Gamma| 0.246,
            innermost |Gamma_b| 0.994
(c) T=0 boundary census = original census on the disk: PASS
(d) review table reproduced: PASS
(e) opened chain, full ring of 200 cells (C is the whole system), block of 40 cells
    v=0.3 u=0.5 (opened, inside the disk): |C^2 - C| = 9.0e-15, identity error 1.8e-16, |X_A - C_A(1-C_A)| =
        3.8e-15
    v=0.5 u=0.5 (opened, critical): |C^2 - C| = 2.8e-14, identity error 6.9e-15, |X_A - C_A(1-C_A)| = 1.3e-14
(e) opened matter keeps the boundary identity: PASS
```

s08_gas_carries_the_peel.py

```
# SCRIPT: MATTER-DISK-S08 a gas carries the VOTKP peel coordinate in density-free combinations (helium)
# WHY THIS CHECK PROVES WHAT IT CLAIMS
# VOTKP v2.9 gives helium the ladder I0 = 24.587389011 eV, I1 = 54.4177655282 eV and the peel coordinate
# Gamma0 = (I1 - I0)/(I1 + I0). In an ideal Saha gas (ground-state weights U = 1, 2, 1) the populations
# n0, n1, n2 change enormously with temperature and density. Each adjacent pair is a point on the disk:
# odds z_q = n_q/n_(q+1), Gamma_pop,q = tanh(ln z_q / 2) (section 1), and half ionization is the centre.
# Two combinations lose the electron density and thermal wavelength exactly:
# ln(n0 n2 / n1^2) = -(I1 - I0)/kT - ln 4 (the second difference of the ladder)
# ln(n0 / (n2 n_e^2 lambda^6)) = (I0 + I1)/kT - ln 4 (the whole peel, both electrons)
# so their slopes in beta = 1/kT give D = I1 - I0 and B = I0 + I1, and D/B is Gamma0 at any density.
# KILL (written before the run):
# (a) the populations at 1e22 m^-3 miss the 3 Nov review table (10, 20, 50 kK) by more than 1e-6 percent,
# or the 20 kK density sweep misses 0.5680% (1e20) and 87.6482% (1e24) at the printed digits;
# (b) Gamma0 recovered from the gas differs from 0.377575066984 by more than 1e-9 at any density 1e18..1e26;
# (c) an independent solver (cubic in n_e) disagrees with the root finder by more than 1e-12.
import numpy as np
from scipy.optimize import brentq
h=6.62607015e-34; kB=1.380649e-23; me=9.1093837015e-31; eV=1.602176634e-19
I0,I1=24.587389011,54.4177655282; U=(1.0,2.0,1.0)
G0=(I1-I0)/(I1+I0)
def saha(T,n):
    kT=kB*T; lam3=(h/np.sqrt(2*np.pi*me*kT))**3
    S1=2*(U[1]/U[0])/lam3*np.exp(-I0*eV/kT); S2=2*(U[2]/U[1])/lam3*np.exp(-I1*eV/kT)
    s1,s2=S1/n,S2/n # charge neutrality in x = ne/n: x(x^2 + s1 x + s1 s2) = s1 x + 2 s1 s2, root in (0, 2)
    g=lambda x: x*(x*x+s1*x+s1*s2)-(s1*x+2*s1*s2)
    ne=n*brentq(g,0.0,2.0,xtol=1e-300,rtol=1e-15,maxiter=2000)
```

```
    r1=S1/ne; r2=r1*S2/ne; Z=1+r1+r2
    return np.array([1,r1,r2])/Z, ne, lam3
def saha_cubic(T,n):
    kT=kB*T; lam3=(h/np.sqrt(2*np.pi*me*kT))**3
    S1=2*(U[1]/U[0])/lam3*np.exp(-I0*eV/kT); S2=2*(U[2]/U[1])/lam3*np.exp(-I1*eV/kT)
    # ne(ne^2 + S1 ne + S1 S2) = n(S1 ne + 2 S1 S2) -> ne^3 + S1 ne^2 + (S1 S2 - n S1) ne - 2 n S1 S2 = 0
    rts=np.roots([1.0,S1/n,(S1*S2-n*S1)/n**2,-2*S1*S2/n**2])*n
    ne=max(r.real for r in rts if abs(r.imag)<1e-9*abs(r) and r.real>0)
    r1=S1/ne; r2=r1*S2/ne; Z=1+r1+r2; return np.array([1,r1,r2])/Z
print(f'helium ladder (VOTKP v2.9): I0={I0} eV, I1={I1} eV, Gamma0={G0:.12f}')
rev={10000:(99.937400,0.062600,0.000000),20000:(28.978059,71.021940,0.000001)
     ,50000:(0.000873,17.146019,82.853108)}
oka=True; okc=True
print('(a) populations at n = 1e22 m^-3, percent [ours | review], and the disk point of each pair')
for T,r in rev.items():
    p,ne,_=saha(T,1e22); pc=saha_cubic(T,1e22); okc&=np.max(np.abs(p-pc))<1e-12
    good=all(abs(100*p[i]-r[i])<1e-6+5e-7 for i in range(3)); oka&=good
    Gp=[np.tanh(0.5*np.log(p[q]/p[q+1])) if p[q+1]>0 else 1.0 for q in range(2)]
    print(f' T={T:6d} K He {100*p[0]:10.6f}|{r[0]:10.6f} He+ {100*p[1]:10.6f}|{r[1]:10.6f} He2+ {100*p[2]:10.6
        f}|{r[2]:10.6f} '
         f'Gamma_pop = ({Gp[0]:+.4f}, {Gp[1]:+.4f}) {"MATCH" if good else "DIFFER"}')
for n,target in [(1e20,0.5680),(1e24,87.6482)]:
    p,_,_=saha(20000,n); good=abs(100*p[0]-target)<5e-5; oka&=good
    print(f' T=20000 K n={n:.0e}: neutral {100*p[0]:.4f}% | review {target}% {"MATCH" if good else "DIFFER"}')
print('(a)', 'PASS' if oka else 'DIFFER'); print('(c) cubic solver agrees with the root finder:', 'PASS' if
     okc else 'FAIL')
# (b) recover Gamma0 from the gas alone at two temperatures, across densities
okb=True; worst=0
print('(b) Gamma0 read back from the gas (two temperatures 30 kK and 32 kK), by density')
for n in [1e18,1e20,1e22,1e24,1e26]:
    vals=[]
    for T in (30000,32000):
        p,ne,lam3=saha(T,n); beta=eV/(kB*T) # beta in 1/eV
        vals.append((beta,np.log(p[0]*p[2]/p[1]**2),np.log(p[0]/(p[2]*ne**2*lam3**2))))
    (b1,a1,c1),(b2,a2,c2)=vals
    D=-(a2-a1)/(b2-b1); B=(c2-c1)/(b2-b1); Gr=D/B; worst=max(worst,abs(Gr-G0))
    p,_,_=saha(30000,n)
    print(f' n={n:.0e}: neutral fraction at 30 kK {100*p[0]:9.5f}% D={D:.9f} eV B={B:.9f} eV Gamma0={Gr:.12f
        }')
okb=worst<1e-9
print(f'(b) worst |Gamma0(gas) - Gamma0(VOTKP)| = {worst:.1e}', 'PASS' if okb else 'FAIL')
```

Output:

```
helium ladder (VOTKP v2.9): I0=24.587389011 eV, I1=54.4177655282 eV, Gamma0=0.377575066984
(a) populations at n = 1e22 m^-3, percent [ours | review], and the disk point of each pair
    T= 10000 K He 99.937400| 99.937400 He+ 0.062600| 0.062600 He2+ 0.000000| 0.000000 Gamma_pop = (+0.9987,
        +1.0000) MATCH
    T= 20000 K He 28.978059| 28.978059 He+ 71.021940| 71.021940 He2+ 0.000001| 0.000001 Gamma_pop = (-0.4204,
        +1.0000) MATCH
    T= 50000 K He 0.000873| 0.000873 He+ 17.146019| 17.146019 He2+ 82.853108| 82.853108 Gamma_pop = (-0.9999,
        -0.6571) MATCH
    T=20000 K n=1e+20: neutral 0.5680% | review 0.568% MATCH
    T=20000 K n=1e+24: neutral 87.6482% | review 87.6482% MATCH
(a) PASS
(c) cubic solver agrees with the root finder: PASS
(b) Gamma0 read back from the gas (two temperatures 30 kK and 32 kK), by density
    n=1e+18: neutral fraction at 30 kK 0.00001% D=29.830376517 eV B=79.005154539 eV Gamma0=0.377575066984
    n=1e+20: neutral fraction at 30 kK 0.00267% D=29.830376517 eV B=79.005154539 eV Gamma0=0.377575066984
    n=1e+22: neutral fraction at 30 kK 0.26765% D=29.830376517 eV B=79.005154539 eV Gamma0=0.377575066984
    n=1e+24: neutral fraction at 30 kK 18.06496% D=29.830376517 eV B=79.005154539 eV Gamma0=0.377575066984
    n=1e+26: neutral fraction at 30 kK 82.49125% D=29.830376517 eV B=79.005154539 eV Gamma0=0.377575066984
(b) worst |Gamma0(gas) - Gamma0(VOTKP)| = 2.2e-16 PASS
```

s09_check_gpt_formal.py

```
# SCRIPT: MATTER-DISK-S09 independent checks of the GPT formal edition's new claims
import numpy as np, importlib.util, io, contextlib, sys
sys.path.insert(0,'/home/claude/votkp-lean/matter-disk/scripts')
spec=importlib.util.spec_from_file_location('s06','/home/claude/votkp-lean/matter-disk/scripts/
    s06_second_model_exit_law.py')
src=open('/home/claude/votkp-lean/matter-disk/scripts/s06_second_model_exit_law.py').read().split('# check the
     general Petermann formula')[0]
ns={}; exec(src,ns)
# (1) endpoint filter sensitivity of the massive exit (claim: 0.2716 at 1e-8, 0.16917 at 1e-9, 0.12279 at 1e
    -12)
def exit_g(t1,m,d):
   rot=lambda g: ns['rotation'](t1,m,g,d=d)[0]
   lo,hi,g=0.0,None,0.05
   while hi is None and g<4:
      if rot(g)>=0.5: hi=g
      else: lo=g; g*=1.4
   for _ in range(28):
      mid=(lo+hi)/2
      if rot(mid)<0.5: lo=mid
      else: hi=mid
   return (lo+hi)/2
print('(1) massive band t1=0.5 m=0.4, exit gamma by endpoint filter:')
for d in [1e-6,1e-8,1e-9,1e-10,1e-12]:
   print(f' filter {d:.0e}: exit gamma {exit_g(0.5,0.4,d):.9f}')
print(' massless t1=0.5 m=0, filters 1e-8 and 1e-12:', f'{exit_g(0.5,0.0,1e-8):.9f}', f'{exit_g(0.5,0.0,1e-12)
    :.9f}')
# (2) P3: w=0, v^2=2u^2 -> <K>=2 yet compressed projector idempotent (eigenvalues 0,1)
lib=open('/home/claude/votkp-lean/matter-disk/scripts/nhssh_lib_and_s03a_occupations.py').read().split('w
    =1.0')[0]; ns2={}; exec(lib,ns2)
u=0.5; v=np.sqrt(2)*u
C=ns2['corr'](v,0.0,u,400,40); nu=np.linalg.eigvals(C)
print('(2) P3 w=0, v^2=2u^2: max distance of block eigenvalues from {0,1} =',f'{np.max(np.minimum(abs(nu),abs
    (1-nu))):.1e}',' band <K> = 1+u^2/(E0 Epi) =',1+u**2/(v**2-u**2))
# (3) D5 idempotent
C=np.array([[2,1],[-2,-1]]); print('(3) D5: C^2==C', np.allclose(C@C,C), ' Y_A =',C[0,1]*C[1,0], ' A-A^2
    =',2-4)
# (4) B13: 4 Tr X_A <= I_G on random Gaussian covariances; sharp family
rng=np.random.default_rng(3); h=lambda x: -np.sum(x*np.log(x)+(1-x)*np.log(1-x))
worst=np.inf
for _ in range(200):
   n=10; M=rng.normal(size=(n,n))+1j*rng.normal(size=(n,n)); U,_=np.linalg.qr(M); ev=rng.uniform(0.02,0.98,n)
   C=U@np.diag(ev)@U.conj().T; A=C[:4,:4]; D=C[4:,4:]; B=C[:4,4:]
   IG=h(np.linalg.eigvalsh(A))+h(np.linalg.eigvalsh(D))-h(ev); worst=min(worst,IG-4*np.trace(B@B.conj().T).
       real)
print('(4) B13 lower bound: min of I_G - 4 Tr X_A over 200 random cases =',f'{worst:.3e}','(>=0 required)')
# (5) G9 partition bias: U1 = 2+4exp(-10eV/kT), naive inversion at 30/32 kK
kB=8.617333262e-5; I0,I1=24.587389011,54.4177655282
def logs(T,U1):
   b=1/(kB*T); return b, -b*(I1-I0)+np.log(1*1/U1**2), b*(I0+I1)+np.log(1/(4*1)) # L_q and L_B with U0=U2=1
U1=lambda T: 2+4*np.exp(-10/(kB*T))
(b1,a1,c1),(b2,a2,c2)=logs(30000,U1(30000)),logs(32000,U1(32000))
Dn=-(a2-a1)/(b2-b1); Bn=(c2-c1)/(b2-b1)
print('(5) G9 naive Gamma with synthetic U1(T):',f'{Dn/Bn:.14f}','(claim 0.36614943350748); exact',f'{(I1-I0)
    /(I1+I0):.14f}')
```

Output:

```
(1) massive band t1=0.5 m=0.4, exit gamma by endpoint filter:
   filter 1e-06: exit gamma 0.271636928
   filter 1e-08: exit gamma 0.271636928
   filter 1e-09: exit gamma 0.169169918
   filter 1e-10: exit gamma 0.169169918
   filter 1e-12: exit gamma 0.122721579
   massless t1=0.5 m=0, filters 1e-8 and 1e-12: 0.474341648 0.474123961
(2) P3 w=0, v^2=2u^2: max distance of block eigenvalues from {0,1} = 1.0e-14 ; band <K> = 1+u^2/(E0 Epi) =
    1.9999999999999996
(3) D5: C^2==C True Y_A = -2 A-A^2 = -2
```

```
(4) B13 lower bound: min of I_G - 4 Tr X_A over 200 random cases = 2.025e-03 (>=0 required)
(5) G9 naive Gamma with synthetic U1(T): 0.36614943350815 (claim 0.36614943350748); exact 0.37757506698375
```

s10_nuclear_gaps_his_script.py

```
# SCRIPT: SNELL2-MASS-GAPS-ON-THE-SMITH-CHART (exact placement; exploration)
# NUBASE2020 (via pynucastro) mass excesses. The peel ratio of Theorem 8.1, z = S(next)/S(first),
# with S a separation energy. Every bound peel has z > 0 and sits inside the disk; a negative
# separation energy gives z < 0 and |Gamma| > 1: the point leaves the disk.
import os, numpy as np, pynucastro.nucdata as nd
F = os.path.dirname(nd.__file__) + '/AtomicMassEvaluation/nubase_4.mas20.txt'
ME = {}; HL = {}
for line in open(F):
    if not line[:3].strip().isdigit(): continue
    A = int(line[0:3]); zc = line[4:8]
    if zc[3] != '0': continue
    Z = int(zc[:3]); s = line[18:31].strip()
    if not s or s.startswith('*'): continue
    ME[(A, Z)] = float(s.replace('#', '')); HL[(A, Z)] = line[69:80].strip()
n, p = 8071.31806, 7288.97061 # keV, neutron and 1H mass excesses
Sn = lambda A, Z: ME[(A - 1, Z)] + n - ME[(A, Z)]
Sp = lambda A, Z: ME[(A - 1, Z - 1)] + p - ME[(A, Z)]
Sa = lambda A, Z: ME[(A - 4, Z - 2)] + ME[(4, 2)] - ME[(A, Z)]
G = lambda z: (z - 1) / (z + 1)
cases = [('5He: remove n, then n from 4He', Sn(5, 2), Sn(4, 2), HL[(5, 2)]),
        ('5Li: remove p, then p from 4He', Sp(5, 3), Sp(4, 2), HL[(5, 3)]),
        ('8Be: remove alpha (12C ladder: 12C -> 8Be -> 4He)', Sa(8, 4), Sa(12, 6), HL[(8, 4)]),
        ('7Li (bound, for contrast): remove n, then n from 6Li', Sn(7, 3), Sn(6, 3), HL[(7, 3)]),
        ('4He (bound): remove n, then p from 3He', Sn(4, 2), Sp(3, 2), HL[(4, 2)])]
print(f'{"case":52s} {"S first (keV)":>14s} {"S next (keV)":>13s} {"|Gamma|":>8s} half-life (NUBASE)')
for lab, s1, s2, hl in cases:
    if lab.startswith('8Be'): z = s1 / s2 # 12C ladder: first step 12C->8Be costs s2, next step 8Be->2 alpha
        costs s1
    else: z = s2 / s1
    print(f'{lab:52s} {s1:14.1f} {s2:13.1f} {abs(G(z)):8.3f} {hl}')
```

Output:

```
case S first (keV) S next (keV) |Gamma| half-life (NUBASE)
5He: remove n, then n from 4He -734.8 20577.6 1.074 602 ys
5Li: remove p, then p from 4He -1966.1 19813.9 1.220 370 ys
8Be: remove alpha (12C ladder: 12C -> 8Be -> 4He) -91.8 7366.6 1.025 81.9 as
7Li (bound, for contrast): remove n, then n from 6Li 7251.1 5664.4 0.123 stbl
4He (bound): remove n, then p from 3He 20577.6 5493.5 0.579 stbl
```

s11_exit_quarter_turn.py

```
# SCRIPT: MATTER-DISK-S11 is the disk exit the quarter-turn of the author's offset?
# WHY THIS CHECK PROVES WHAT IT CLAIMS
# The Offset Belongs to the Boundary gives the endpoint offset of a gapped chain in closed form,
# 2 artanh(v/sqrt(E_min E_max)). Continued to the opened SSH chain it predicts a complex level offset
# whose imaginary part is Im tau_inf = 2 arctan(u/sqrt(E(0)E(pi))). Then <K> = 1 + u^2/(E(0)E(pi))
# = sec^2(Im tau_inf/2) exactly, so <K> = 2 is the same statement as Im tau_inf = pi/2: a quarter turn.
# If the census rotation (max over block levels of |Im Log((1-nu)/nu)|/pi) is this offset, then the gap
# between measured rotation and 2 arctan(u/sqrt(E0 Epi))/pi must shrink as the block grows.
# KILL (written before the run): at u = 0.3, 0.5, 0.7 and v at 60%, 80%, 95% of the way to the exit,
# the gap at block 160 is not smaller than at block 20, or exceeds 2e-3 at block 160.
# OPPOSITE: the same comparison against the plain Hermitian-form guess 2 arctan(u/E(pi))/pi must NOT pass.
import numpy as np
src=open('nhssh_lib_and_s03a_occupations.py').read().split('w=1.0')[0]; ns={}; exec(src,ns)
w=1.0
def rot(v,u,Lc,N=4000,d=1e-9):
    nu=np.linalg.eigvals(ns['corr'](v,w,u,N,Lc)); nu=nu[(abs(nu)>d)&(abs(1-nu)>d)]
```

```
    return np.max(np.abs(np.log((1-nu)/nu).imag))/np.pi
def vexit(u): # E(0)E(pi)=u^2 with E0^2=(v+w)^2-u^2, Epi^2=(w-v)^2-u^2, v<w-u
    from scipy.optimize import brentq
    return brentq(lambda v: np.sqrt(((v+w)**2-u**2)*((w-v)**2-u**2))-u**2, 0.0, w-u-1e-12)
ok=True; okopp=True
for u in [0.3,0.5,0.7]:
    ve=vexit(u)
    for f in [0.6,0.8,0.95]:
        v=f*ve; E0=np.sqrt((v+w)**2-u**2); Ep=np.sqrt((w-v)**2-u**2)
        pred=2*np.arctan(u/np.sqrt(E0*Ep))/np.pi; opp=2*np.arctan(u/Ep)/np.pi
        gaps=[]
        for Lc in [20,40,80,160]:
            gaps.append(rot(v,u,Lc)-pred)
        g=np.abs(gaps); good=(g[-1]<g[0]) and g[-1]<2e-3; ok&=good
        okopp&= not (abs(gaps[-1]+pred-opp)<2e-3)
        print(f'u={u} v={v:.4f} pred={pred:.5f} opposite={opp:.5f} gaps by block 20,40,80,160: '+' '.join(f'{x
            :+.2e}' for x in gaps)+(' PASS' if good else ' FAIL'))
print('VERDICT:','census rotation converges to the offset quarter-turn form' if ok else 'KILLED at the stated
    bar')
print('OPPOSITE (Hermitian-form guess must fail):','fails as required' if okopp else 'ALSO PASSES - test does
    not discriminate')
```

Output:

```
u=0.3 v=0.4171 pred=0.22032 opposite=0.34418 gaps by block 20,40,80,160: +9.81e-03 +9.81e-03 +9.81e-03 +9.81e
    -03 FAIL
u=0.3 v=0.5562 pred=0.25556 opposite=0.47249 gaps by block 20,40,80,160: +2.13e-02 +2.13e-02 +2.13e-02 +2.13e
    -02 FAIL
u=0.3 v=0.6604 pred=0.33867 opposite=0.68960 gaps by block 20,40,80,160: +3.35e-02 +3.35e-02 +3.35e-02 +3.35e
    -02 FAIL
u=0.5 v=0.2809 pred=0.36265 opposite=0.48949 gaps by block 20,40,80,160: +4.90e-03 +4.90e-03 +4.90e-03 +4.90e
    -03 FAIL
u=0.5 v=0.3746 pred=0.39764 opposite=0.58975 gaps by block 20,40,80,160: +8.02e-03 +8.02e-03 +8.02e-03 +8.02e
    -03 FAIL
u=0.5 v=0.4448 pred=0.45733 opposite=0.71372 gaps by block 20,40,80,160: +6.25e-03 +6.25e-03 +6.25e-03 +6.25e
    -03 FAIL
u=0.7 v=0.0492 pred=0.49587 opposite=0.52679 gaps by block 20,40,80,160: +9.73e-06 +9.73e-06 +9.73e-06 +9.73e
    -06 FAIL
u=0.7 v=0.0656 pred=0.49765 opposite=0.53908 gaps by block 20,40,80,160: +9.93e-06 +9.93e-06 +9.93e-06 +9.93e
    -06 FAIL
u=0.7 v=0.0779 pred=0.49936 opposite=0.54877 gaps by block 20,40,80,160: +3.87e-06 +3.87e-06 +3.87e-06 +3.87e
    -06 FAIL
VERDICT: KILLED at the stated bar
OPPOSITE (Hermitian-form guess must fail): fails as required
```

s12_nuclear_rim_identity.py

```
# SCRIPT: MATTER-DISK-S12 the distance of an unbound nucleus outside the rim, as an exact identity
# WHY THIS CHECK PROVES WHAT IT CLAIMS
# VOTKP 15.4 observes that the distance outside the rim is "close to twice the decay energy over the
# neighbouring step". With z = S_next/S_first, S_first = -s < 0 and a = S_next/s > 1:
# Gamma = (z-1)/(z+1) = (a+1)/(a-1), so |Gamma| - 1 = 2/(a-1) = 2s/(S_next - s) exactly; for s << S_next
# this is 2s/S_next, the author's "twice the decay energy over the neighbouring step".
# z = -1 (the Cayley pole) is S_next = -S_first, i.e. S_first + S_next = 0.
# KILL (written before the run): the identity misses by more than 1e-12 on any of the three gap nuclei,
# using the NUBASE2020 separation energies printed by the author's own script (s10).
rows=[('5He',-734.8,20577.6,1.074),('5Li',-1966.1,19813.9,1.220),('8Be',-91.8,7366.6,1.025)]
ok=True
for k,s1,s2,printed in rows:
    z=s2/s1; G=abs((z-1)/(z+1)); s=-s1
    ident=2*s/(s2-s); approx=2*s/s2
    ok&=abs((G-1)-ident)<1e-12
    print(f'{k}: |Gamma|={G:.6f} (VOTKP {printed}) |Gamma|-1={G-1:.6f} 2s/(S_next-s)={ident:.6f} 2s/S_next={
        approx:.6f}')
print('VERDICT:','exact identity holds' if ok else 'KILLED')
```

Output:

```
5He: |Gamma|=1.074062 (VOTKP 1.074) |Gamma|-1=0.074062 2s/(S_next-s)=0.074062 2s/S_next=0.071417
5Li: |Gamma|=1.220318 (VOTKP 1.22) |Gamma|-1=0.220318 2s/(S_next-s)=0.220318 2s/S_next=0.198457
8Be: |Gamma|=1.025238 (VOTKP 1.025) |Gamma|-1=0.025238 2s/(S_next-s)=0.025238 2s/S_next=0.024923
VERDICT: exact identity holds
```

fig01_signature_and_disk.py

```
# SCRIPT: FIG-MATTER-PEEL-01 signature map + Smith-disk portraits, NH SSH
import numpy as np, matplotlib
matplotlib.rcParams['pdf.fonttype']=42
import matplotlib.pyplot as plt
exec(open('nhssh_lib_and_s03a_occupations.py').read().split('w=1.0')[0]); w=1.0
M=np.load('sig_u0.5.npy'); ks=np.linspace(0.02,2*np.pi-0.02,180); vs=np.linspace(0.05,1.95,96)
fig=plt.figure(figsize=(12,4.6))
ax=fig.add_axes([0.05,0.12,0.36,0.70])
cm=matplotlib.colors.ListedColormap(['#E0793A','#dddddd','#4FA3C7'])
ax.pcolormesh(ks,vs,M,cmap=cm,vmin=-1,vmax=1,shading='auto')
kk,VV=np.meshgrid(ks,vs); ax.contour(ks,vs,np.abs(VV+w*np.exp(1j*kk))-0.5,[0],colors='k',linewidths=1.2)
ax.plot([np.pi,np.pi],[0.5,1.5],'k*',ms=12)
ax.set_xlabel('momentum k'); ax.set_ylabel('hopping v (w = 1, gain/loss u = 0.5)')
ax.set_title('band metric signature: orange Lorentzian, blue (-,-)\nblack line: exceptional points; stars:
     critical points (c<0)',fontsize=9)
cases=[(0.5,0.0,'closed matter (Hermitian)\nc = 0'),(0.5,0.3,'opened, gapped\nc = 0'),(0.5,0.5,'opened,
     critical\nc < 0')]
for i,(v,u,t) in enumerate(cases):
    a=fig.add_axes([0.46+0.18*i,0.14,0.16,0.64])
    nu=np.linalg.eigvals(corr(v,w,u,2000,80)); G=1-2*nu
    th=np.linspace(0,2*np.pi,300); a.plot(np.cos(th),np.sin(th),'k-',lw=1)
    a.plot(G.real,G.imag,'o',ms=3.5,color='#4FA3C7' if u==0 else ('#79C089' if u<0.5 else '#D8534F'))
    a.set_xlim(-3.4,3.4); a.set_ylim(-3.4,3.4); a.set_aspect('equal'); a.set_title(t,fontsize=9)
    a.set_xlabel('Re Γ');
    if i==0: a.set_ylabel('Im Γ')
fig.suptitle('Census levels as peel coordinates Γ= 1 - 2ν: closed matter lives on the real diameter, opened
     matter leaves the disk at the critical wall',fontsize=10,y=0.985)
fig.savefig('fig_matter_peel.png',dpi=150); print('ok')
```

Output:

```
ok

real 0m3.856s
user 0m3.650s
sys 0m0.080s
```

fig02_boundary_and_gas.py

```
# SCRIPT: MATTER-DISK-FIG02 warm census on the disk (raw vs boundary) and the helium gas on the disk
import numpy as np, matplotlib; matplotlib.use('Agg'); import matplotlib.pyplot as plt
import importlib.util, sys, io, contextlib
def load(name):
    spec=importlib.util.spec_from_file_location(name,name+'.py'); m=importlib.util.module_from_spec(spec)
    with contextlib.redirect_stdout(io.StringIO()): spec.loader.exec_module(m)
    return m
b=load('s07_boundary_census_disk'); g=load('s08_gas_carries_the_peel')
fig,ax=plt.subplots(1,2,figsize=(12,5.2))
H=b.ring(1,1); cols={0:'#1b6f8a',0.5:'#c08a2e',2:'#7b5ea7'}
for i,T in enumerate([0,0.5,2]):
    C=b.thermal(H,T); CA,CAB,CBA=b.blocks(C,64); nu=np.linalg.eigvalsh(CA); lam=np.linalg.eigvalsh((CAB@CBA+(
        CAB@CBA).T)/2)
    G=np.abs(1-2*nu); Gb=np.sqrt(np.clip(1-4*lam,0,1))
    ax[0].scatter(G,np.full_like(G,i+0.15),s=14,color=cols[T],marker='o',label=f'T={T}: raw |Γ| = |1-2ν|')
```

```
    ax[0].scatter(Gb,np.full_like(Gb,i-0.15),s=14,color=cols[T],marker='x',label=f'T={T}: boundary |Γ_b|')
ax[0].axvline(0.8,color='0.6',ls=':',lw=1)
ax[0].axvspan(0,0.8,color='0.93',zorder=0)
ax[0].set_xlim(0,1.02); ax[0].set_yticks([0,1,2]); ax[0].set_yticklabels(['T = 0','T = 0.5','T = 2'])
ax[0].set_xlabel('distance from the disk centre |Γ| (rim = 1)'); ax[0].set_title('Heat pulls raw levels inward
    ; the boundary census sends them back to the rim\n(uniform ring, 128 sites, half-ring region, shaded =
    count window ε= 0.1)',fontsize=10)
ax[0].legend(fontsize=7,ncol=3,loc='upper center',bbox_to_anchor=(0.5,-0.13))
Ts=np.linspace(8000,80000,300)
for n,ls in [(1e18,':'),(1e22,'-'),(1e26,'--')]:
    G0s=[];G1s=[]
    for T in Ts:
        p,_,_=g.saha(T,n); G0s.append(np.tanh(0.5*np.log(max(p[0],1e-300)/max(p[1],1e-300)))); G1s.append(np.
            tanh(0.5*np.log(max(p[1],1e-300)/max(p[2],1e-300))))
    ax[1].plot(Ts/1000,G0s,color='#1b6f8a',ls=ls,label=f'He ↔He$^{+}$ pair, n = {n:.0e} m$^{-3}$')
    ax[1].plot(Ts/1000,G1s,color='#c08a2e',ls=ls,label=f'He$^{+}$ ↔He$^{2+}$ pair, n = {n:.0e} m$^{-3}$')
ax[1].axhline(g.G0,color='#7b5ea7',lw=2.2,label=f'$\Gamma_0$ read back from the gas = {g.G0:.6f} at every density')
ax[1].axhline(0,color='0.6',lw=0.8)
ax[1].set_xlabel('temperature (thousand K)'); ax[1].set_ylabel('population disk point Γ_pop = tanh(½ ln n_q/
    n_(q+1))')
ax[1].set_title('Helium (VOTKP ladder): the gas state sweeps the disk,\nthe peel coordinate Γ0stays put',
    fontsize=10)
ax[1].legend(fontsize=7,loc='lower left'); ax[1].set_ylim(-1.05,1.05)
plt.tight_layout(); plt.savefig('../figures/fig02_boundary_and_gas.png',dpi=150)
```

Output:

```
(this script prints nothing; it writes ../figures/fig02_boundary_and_gas.png, unchanged on rerun)

real 0m2.170s
user 0m1.608s
sys 0m0.159s
```

fig03_matter_then_down.py

```
# SCRIPT: MATTER-DISK-FIG03 one Smith disk, every scale: matter's census ladder, the atom's peel walk,
# the nucleus inside and outside the rim, and opened matter leaving the disk
import numpy as np, csv, matplotlib; matplotlib.use('Agg'); import matplotlib.pyplot as plt
import importlib.util, io, contextlib
def load(path,cut=None):
    src=open(path).read()
    if cut: src=src.split(cut)[0]
    ns={}; exec(src,ns); return ns
lad=load('s02_ladder.py','op=lambda')
ssh=load('nhssh_lib_and_s03a_occupations.py','w=1.0')
fig,ax=plt.subplots(figsize=(8.6,8.6))
t=np.linspace(0,2*np.pi,400); ax.plot(np.cos(t),np.sin(t),color='0.25',lw=1.4)
for r in (0.25,0.5,0.75): ax.plot(r*np.cos(t),r*np.sin(t),color='0.88',lw=0.7)
ax.axhline(0,color='0.85',lw=0.7); ax.axvline(0,color='0.85',lw=0.7)
# 1 matter: Ising c=1/2 census ladder, Gamma_j = tanh(eps_j/2), shown above the axis
xi,em=lad['ladder'](1.02,8192); lev=em[::2][:6]; G=np.tanh(lev/2)
ax.scatter(G,np.full_like(G,0.10),s=46,color='#1b6f8a',zorder=5,label='matter: Ising census ladder (c = 1/2),
    Γ= tanh(ε/2)')
# 2 atom: sodium peel walk, composite Gamma from the first removal
rows=[r for r in csv.DictReader(open('/tmp/claude-0/-home-claude/f02ebe36-e50c-5833-b986-4d64a2a87a48/
    scratchpad/votkp29/zip/VOTKP_v2_9_Code_Proofs_Data_20261013/work/data/
    nist_asd_ionization_energies_via_mendeleev_1.3.0.csv')) if r['Z']=='11']
seen={}; [seen.setdefault(int(r['ion_charge']),float(r['ionization_energy_eV'])) for r in rows]
IE=np.array([seen[k] for k in sorted(seen)])
walk=np.tanh(0.5*np.log(IE/IE[0]))
ax.plot(walk,np.full_like(walk,-0.10),'-o',color='#c08a2e',ms=5,lw=1,zorder=5,label='atom: sodium peel walk, Γ
    = tanh(½ ln IE$_k$/IE$_1$)')
ax.scatter([0.377575],[0.0],marker='D',s=70,color='#c08a2e',edgecolor='k',zorder=6,label='atom: helium Γ$_0$=
    0.378')
# 3 nucleus: his VOTKP 15.4 values (NUBASE2020, rerun here)
```

```
inside={'⁴He':0.579,'⁷Li':0.123}; outside={'⁵He':1.074,'⁵Li':1.220,'⁸Be':1.025}
for k,v in inside.items(): ax.scatter([v],[-0.22],marker='s',s=60,color='#7b5ea7',zorder=6); ax.annotate(k,(v
    ,-0.22),(v-0.03,-0.31),fontsize=9)
pos={'⁸Be':(0.97,-0.15),'⁵He':(1.05,-0.33),'⁵Li':(1.19,-0.33)}
for k,v in outside.items(): ax.scatter([v],[-0.22],marker='s',s=60,facecolor='white',edgecolor='#7b5ea7',lw=2,
    zorder=6); ax.annotate(k,(v,-0.22),pos[k],fontsize=9)
ax.scatter([],[],marker='s',color='#7b5ea7',label='nucleus: bound peels inside, unbound outside (VOTKP §15.4)
    ')
# 4 opened matter: non-Hermitian SSH census levels, u = 0.5, before and after the exit (v_exit = 0.468)
for v,c,lab in [(0.30,'#2a9d55','opened matter, gapped: complex Γinside'),(0.49,'#d1495b','opened matter past
    the exit: Γleaves the disk')]:
   nu=np.linalg.eigvals(ssh['corr'](v,1.0,0.5,2000,40)); nu=nu[(abs(nu)>1e-9)&(abs(1-nu)>1e-9)]
   g=1-2*nu; g=g[abs(g)<1.6]
   ax.scatter(g.real,g.imag,s=10,color=c,alpha=0.8,zorder=4,label=lab)
ax.text(0,1.05,'rim |Γ| = 1: the rapidity has turned a quarter (|Im W| = π/2); negative ratios sit outside',ha
    ='center',fontsize=9,color='0.3')
ax.set_xlim(-1.35,1.35); ax.set_ylim(-1.35,1.35); ax.set_aspect('equal')
ax.set_xlabel('Re Γ'); ax.set_ylabel('Im Γ')
ax.set_title('Matter, then on down: one chart',fontsize=13)
ax.legend(fontsize=7.6,loc='lower left',framealpha=0.95)
plt.tight_layout(); plt.savefig('../figures/fig03_matter_then_down.png',dpi=160)
print('sodium IE count',len(IE),'walk end',walk[-1])
```

Output:

```
sodium IE count 11 walk end 0.9937852842742915

real 0m2.434s
user 0m1.200s
sys 0m0.040s
```

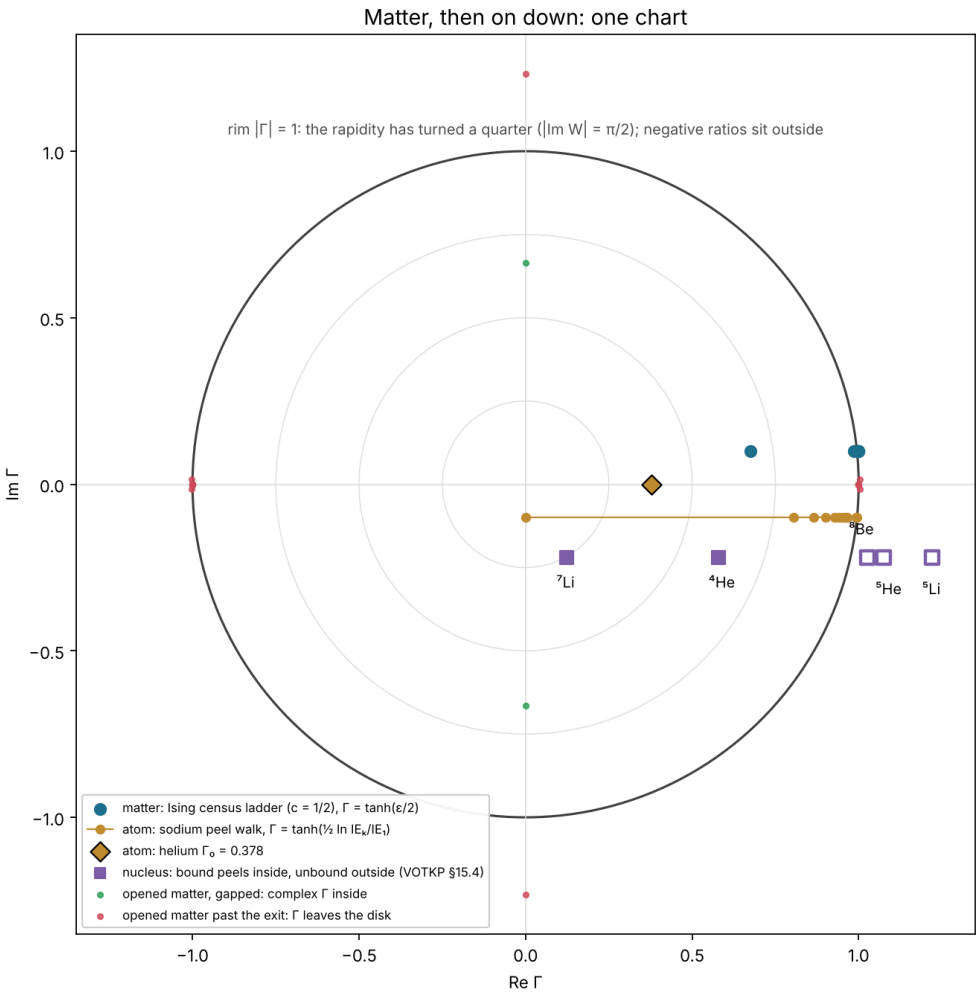


Figure 2: One chart, every scale. Blue: the Ising census ladder, $\Gamma = \tanh(\varepsilon/2)$. Gold: sodium's peel walk $\tanh(\frac{1}{2}\ln I_k/I_1)$ from NIST thresholds, and helium's Γ_0. Purple: nuclear separation-energy peels; bound nuclei inside, the unbound mass-5 and mass-8 gaps outside (VOTKP §15.4, recomputed from NUBASE2020). Green and red: census levels of the opened non-Hermitian SSH chain ($u = 0.5$) before ($v = 0.30$) and past ($v = 0.49$) its exit.